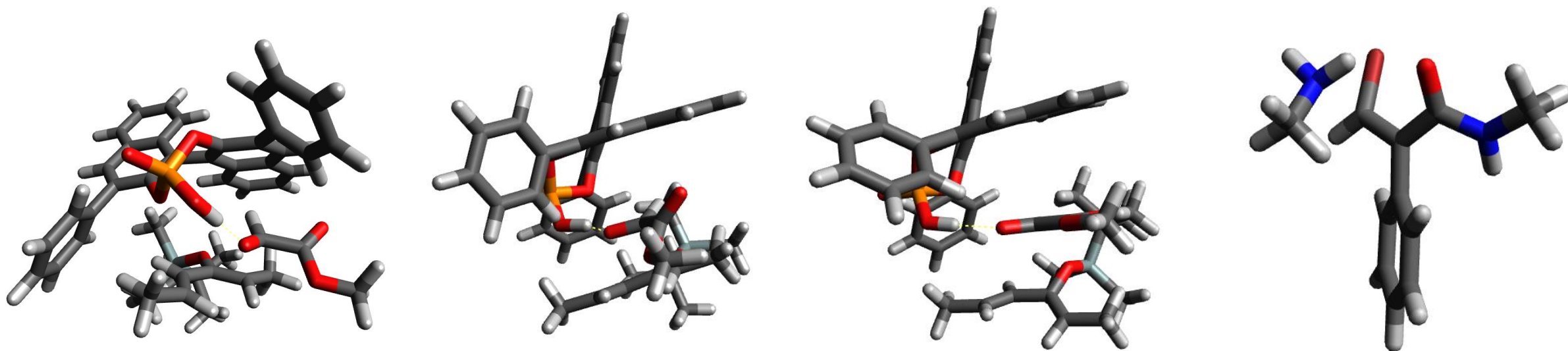
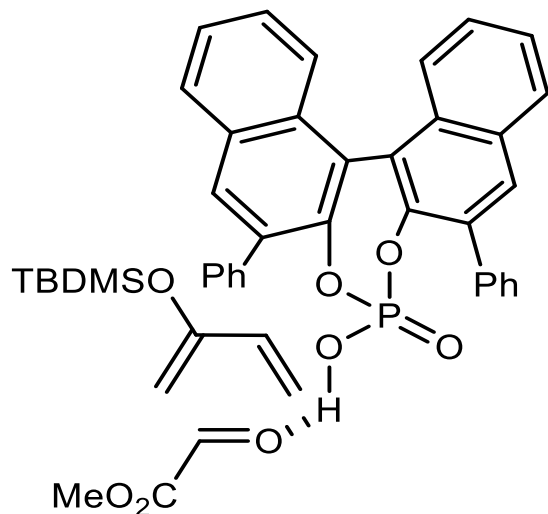


TS searching with mixed method Hessians



January 2024

Pre-optimising at a lower level of theory



BPA_TS_1
(catalysed hetero Diels-Alder)

SQM (PM6 PCM(toluene)):

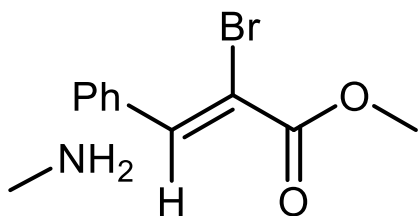
4x optimisation cycles, 153 steps, 11h-17m-47.3s

HF (HF 6-31G* PCM(toluene))

8x optimisation cycles, 374 steps, 15d-14h-22m-54.5s

LDA (SVWN def2-SVP PCM(toluene))

10+ optimisation cycles, 500+ steps, 16d+



TS_68
(uncatalysed aza-Michael)

SQM (PM6 PCM(toluene)):

1x optimisation cycle, 36 steps, 19m-55.3s

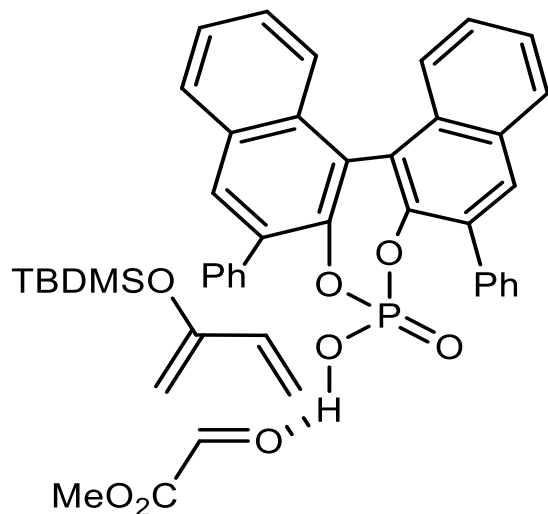
HF (HF 6-31G* PCM(toluene))

1x optimisation cycle, 23 steps, 1h-15m-11.6s

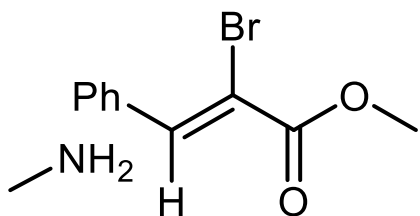
LDA (SVWN def2-SVP PCM(toluene))

3x optimisation cycles, 131 steps, 9h-38m-31.7s

Re-optimising lower-level structures to DFT (wB97XD def2-SVP PCM(toluene))



BPA_TS_1
(catalysed hetero Diels-Alder)



TS_68
(uncatalysed aza-Michael)

SQM $\rightarrow$ DFT :

?x optimisation cycle, ? steps, ?

HF $\rightarrow$ DFT :

?x optimisation cycle, ? steps, ?

LDA $\rightarrow$ DFT :

?x optimisation cycles, ? steps, ?

DFT:

6x optimisation cycles, 73 steps, 17d-8h-17m-43s

SQM $\rightarrow$ DFT :

4x optimisation cycles, 161 steps, 22h-43m-54.8s

HF $\rightarrow$ DFT :

1x optimisation cycle, 16 steps, 1h-59m-55.9s

LDA $\rightarrow$ DFT :

4x optimisation cycles, 181 steps, 21h-59m-30.1s

DFT:

1x optimisation cycle, 27 steps, 4h-11m-38.1s

External keyword for Gaussian interface

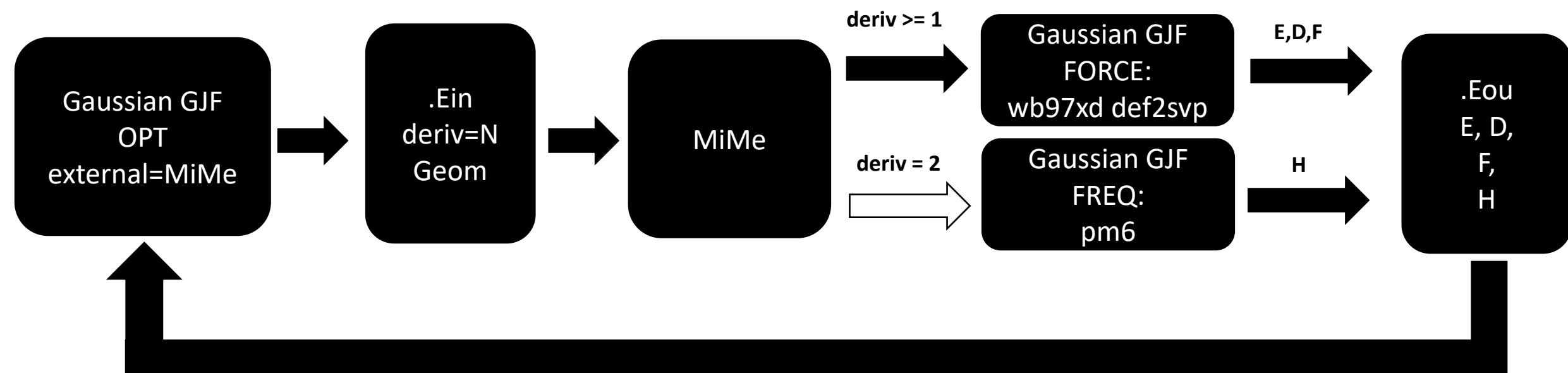
Previously:

1. Run separate Freq=noraman calculation at lower level of theory (e.g. PM6, HF 6-31G*, SVWN def2-SVP) with %chk=freq.chk keyword
2. Move checkpoint file from frequency calculation to folder with optimisation Gaussian input file
3. Use %chk=freq.chk with opt=(readfc,ts,noeigen,maxcycles=N,...) keyword arguments for optimisation
4. This is designed to load in the low-level hessian into a higher-level TS optimisation, with new Hessians recalculated every N steps

External keyword:

1. Run TS optimisation all in one Gaussian input file using:
external="~/scripts/MiMe -H 'pm6' -EF 'wb97xd def2svp' -c 12 -m 48GB" opt=(recalcfc=N,ts,noeigen,...,nomicro)
2. No need for additional frequency calculations, but does require a working script to interface with G16 via External
3. G16 sends the python script (MiMe) a Gau-12345.EIn "input" file containing [a] deriv = d where d is the derivatives of Energy, and [b] the molecule specification (charge, spin and geometry in standard orientation in Bohrs)
4. Python script reads in Gau-12345.EIn and sends off a new G16 calculation of Energy, Force, or Freq (if d = 0,1,2) and returns those values to the main G16 task via the Gau-12345.EOu "output" file

External keyword for Gaussian interface



External keyword for Gaussian interface

1. Currently MiMe does not replicate the results obtained by using an external frequency calculation, reading in the Hessians via the `opt=readfc` keyword argument. The calculations begin very similarly, with the **identical** Forces and **similar** Eigenvalues but these numbers **diverge** the longer the calculation goes on for – leading to slightly more steps being taken with External and therefore taking longer:
2. Currently testing on DA-1_2 (acrolein Diels-Alder dimerization) at **23 steps vs 21 steps** and on TS_68 (uncatalysed aza-Michael reaction) at **42 steps vs 42 steps** for `calcfc`
3. One possibility is that each Freq calculation involves a putative Berny optimisation step, and the Hessian after optimisation is what is used by `readfc` whereas the Hessian before optimisation is used by External (or vice versa). i.e. the Hessians are incorrect from the start (or from the 2nd step onwards)
4. Another explanation is that the Hessians begin the same but are **updated** in different ways by the Berny / LBFGS update scheme over the course of optimisation.
5. It would be worth getting the **External** keyword to work, since this would allow us to intercept the low-level Hessian and improve it with ML before passing it back to G16, and because the process is more self-contained (no explicit Freq calculations).
6. The problem is likely to involve the use of the Hessian and Forces in converting between Cartesian coordinates and Internal Coordinates – when mixed Hessians and Forces are used the change of coordinates introduces an error

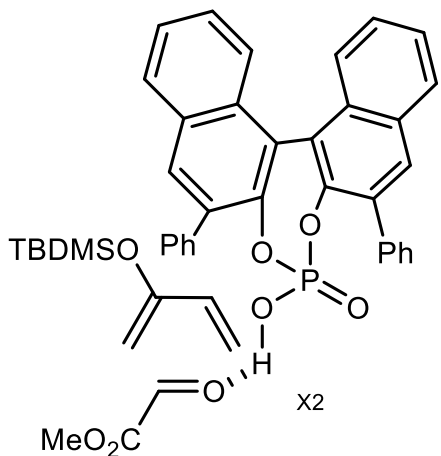
External keyword for Gaussian interface

Once the `external` keyword is able to replicate the optimisation results for `readfc`:

1. Investigate different frequencies of Hessian calculation using `opt=(calcfc,recalcfc=N,...)` for `N = {1=calcall, 2, 5, 10, 20, 50=calcfc}`
2. Create a script to intercept the Geometry and Hessians from MiMe and use delta-ML to improve the low-level Hessian towards the higher-level (DFT) Hessian
3. Test ML Hessians with different combinations of low and high level of theory (or select one high level of theory and try different low levels e.g. SMQ-wB97xd, HF-wB97xd, LDA-wB97xd)

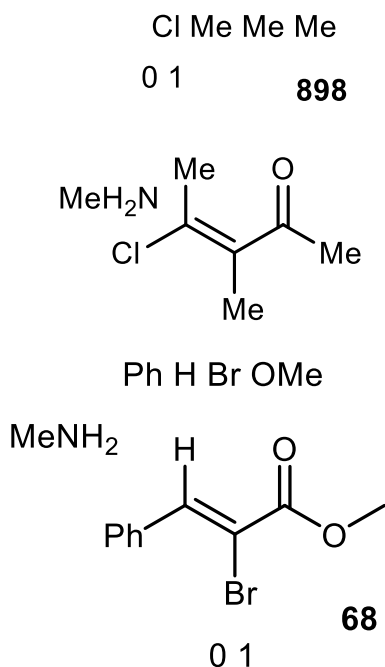
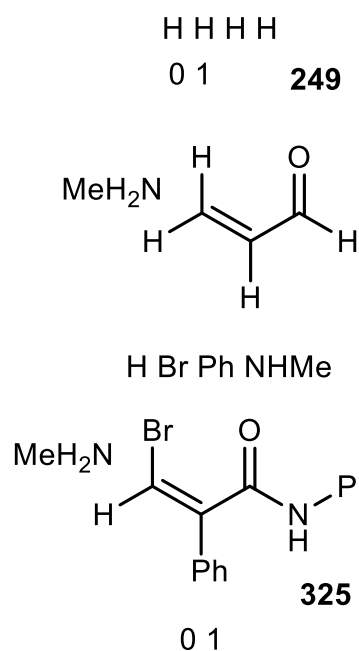
TS structures: OPLS3e initial geometries (reacting bonds constrained)

**Phosphoric-acid catalysed
Hetero Diels-Alder
(two different conformers)**
BPA_1, BPA_2



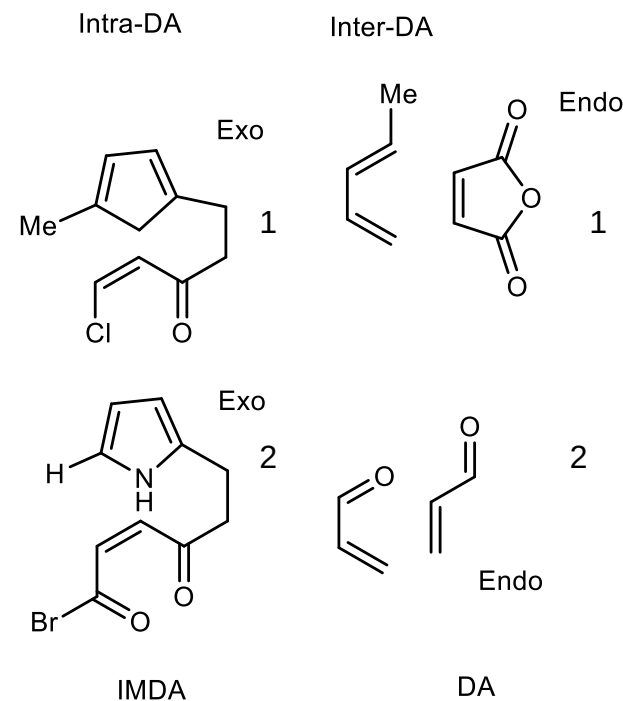
Aza-Michael reactions

AM_249, AM_898,
AM_325, AM_68

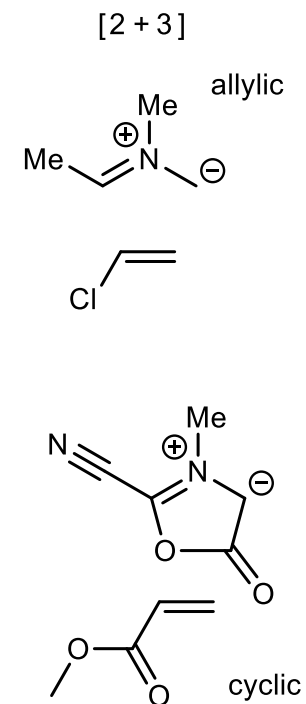


Diels-Alder reactions

IMDA_1, IMDA_2,
DA_1, DA_2



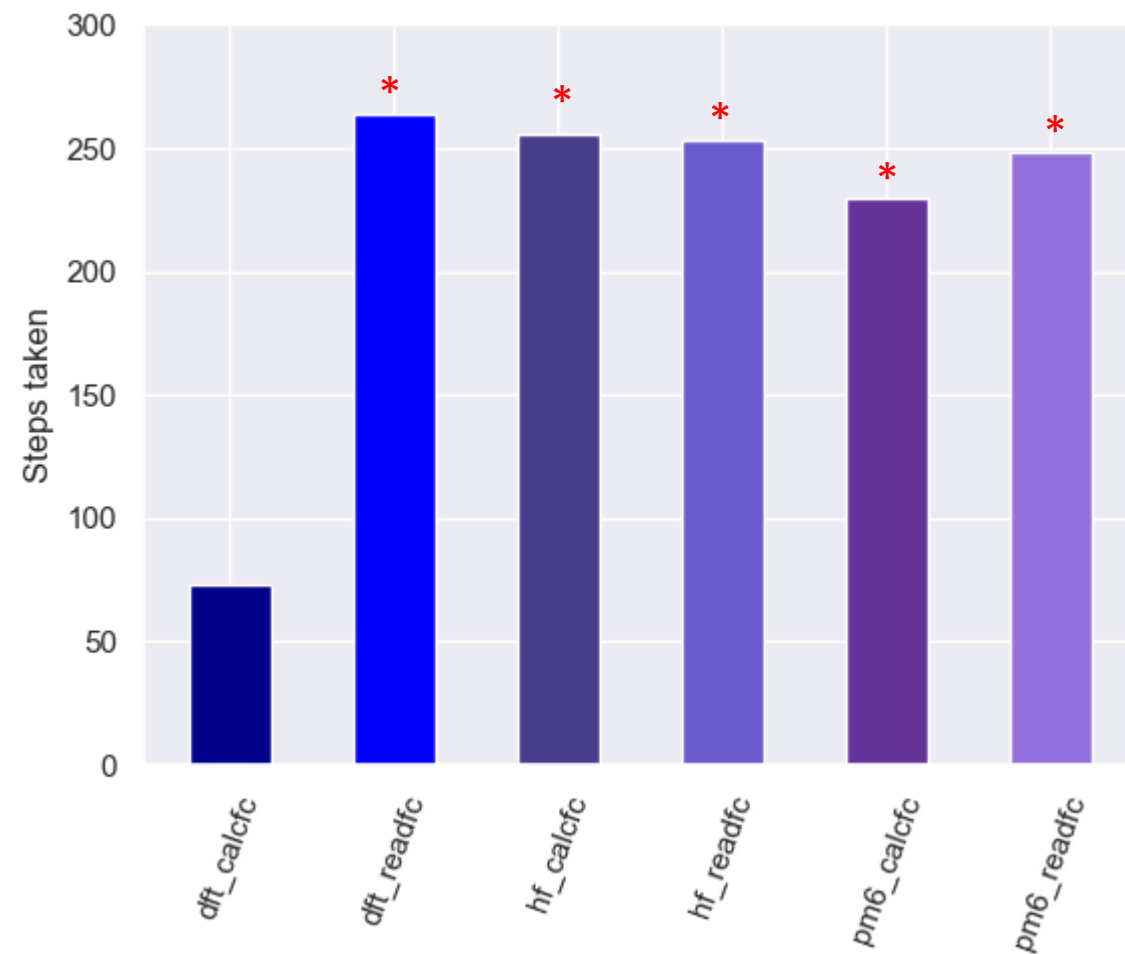
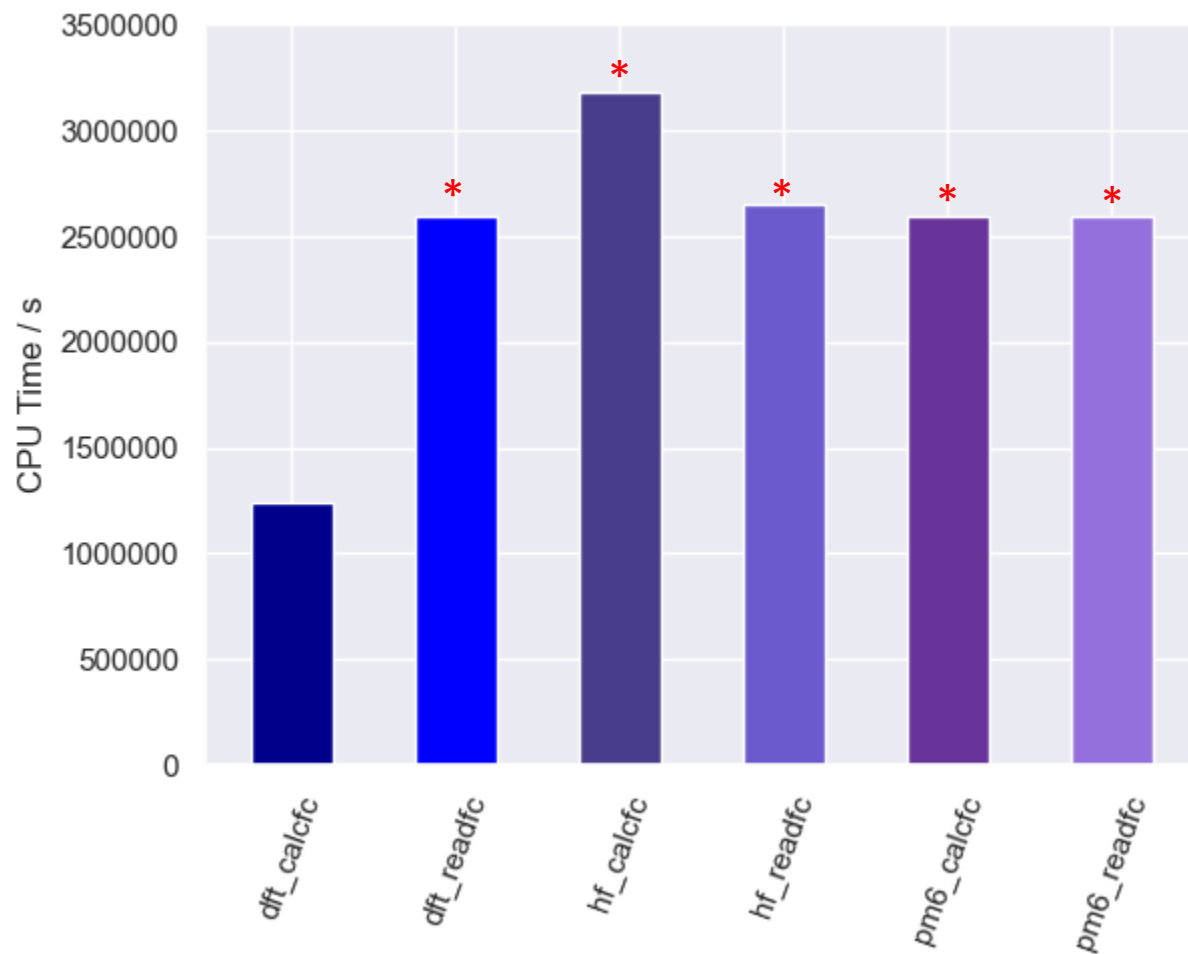
**1,3-Dipolar
cycloadditions**
allylic, cyclic



Hessian levels of theory: SQM, HF, DFT

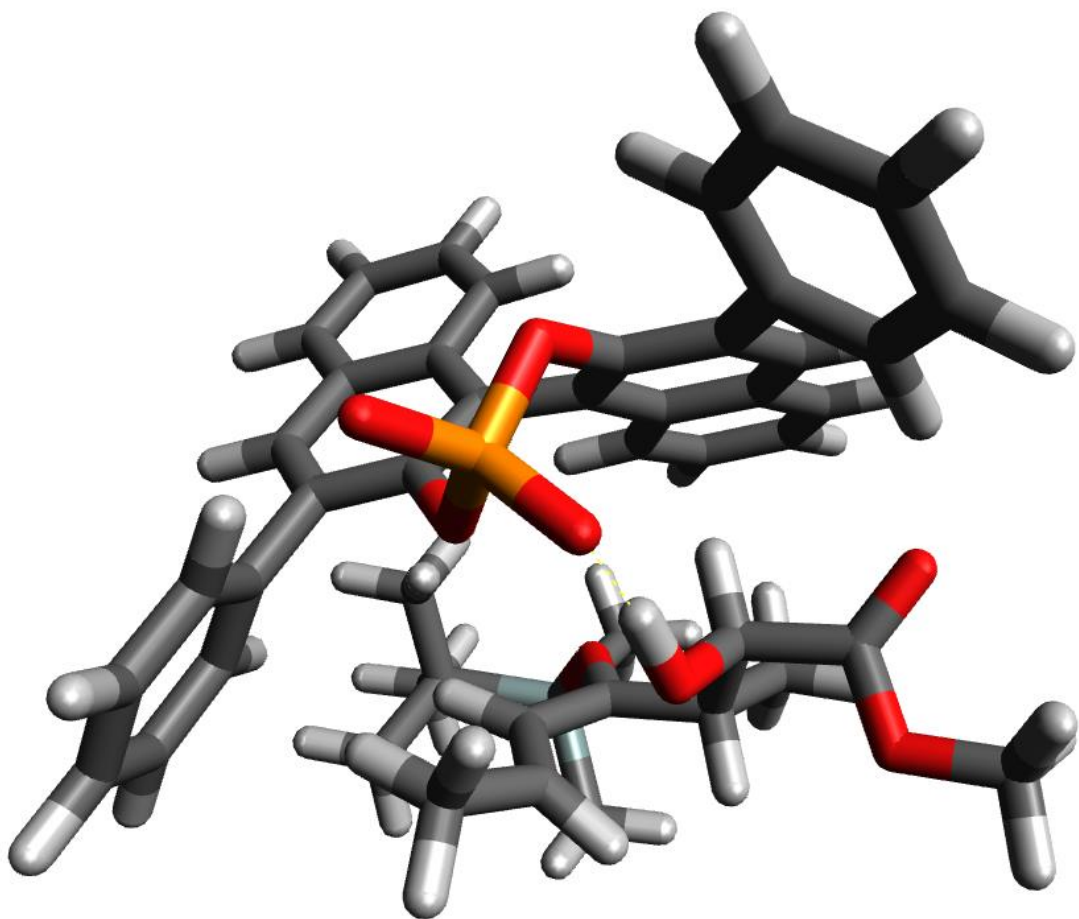
	DFT (wB97xd/def2-SVP)	HF (HF/6-3G(d))	SQM (PM6)
ReadFC	<pre>%chk=BPA_TS_1_opt_dft.chk %nprocshared=12 %mem=48GB # wb97xd def2svp geom=allchk scrf=(iefpcm,solvent=toluene) opt=(readfc,ts,noeigen,maxcycles=50,maxstep=10)</pre>	<pre>%chk=BPA_TS_1_opt_hf.chk %nprocshared=12 %mem=48GB # wb97xd def2svp geom=allchk scrf=(iefpcm,solvent=toluene) opt=(readfc,ts,noeigen,maxcycles=50,maxstep=10) (BPA_TS_1_hf.chk obtained from previous opt)</pre>	<pre>%chk=BPA_TS_1_opt_sqm.chk %nprocshared=12 %mem=48GB # wb97xd def2svp geom=allchk scrf=(iefpcm,solvent=toluene) opt=(readfc,ts,noeigen,maxcycles=50,maxstep=10) (BPA_TS_1_opt_sqm.chk obtained from previous opt)</pre>
CalcFC	<pre>%chk=BPA_TS_1_opt_dft_re.chk %nprocshared=12 %mem=48GB # wb97xd def2svp geom=allchk scrf=(iefpcm,solvent=toluene) opt=(calcfc,ts,noeigen,maxcycles=50,maxstep=10)</pre>	<pre>%chk=BPA_TS_1_opt_hf.chk %nprocshared=12 %mem=48GB # wb97xd def2svp geom=allchk scrf=(iefpcm,solvent=toluene) opt=(readfc,ts,noeigen,maxcycles=50,maxstep=10) (BPA_TS_1_hf.chk obtained from a new hf 6-31g(d) hessian calculation)</pre>	<pre>%chk=BPA_TS_1_opt_sqm.chk %nprocshared=12 %mem=48GB # wb97xd def2svp geom=allchk scrf=(iefpcm,solvent=toluene) opt=(readfc,ts,noeigen,maxcycles=50,maxstep=10) (BPA_TS_1_opt_sqm.chk obtained from a new pm6 hessian calculation)</pre>

BPA_1

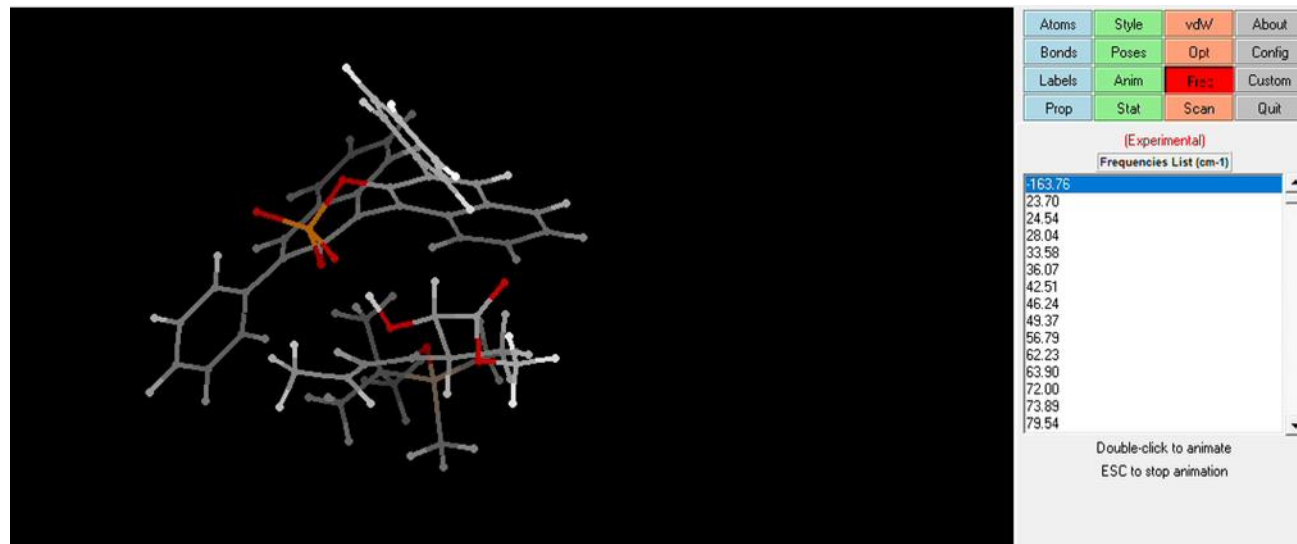


* TS structure did not optimise after 10 job cycles

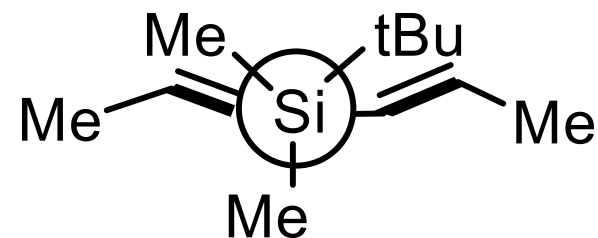
BPA_1: conformation



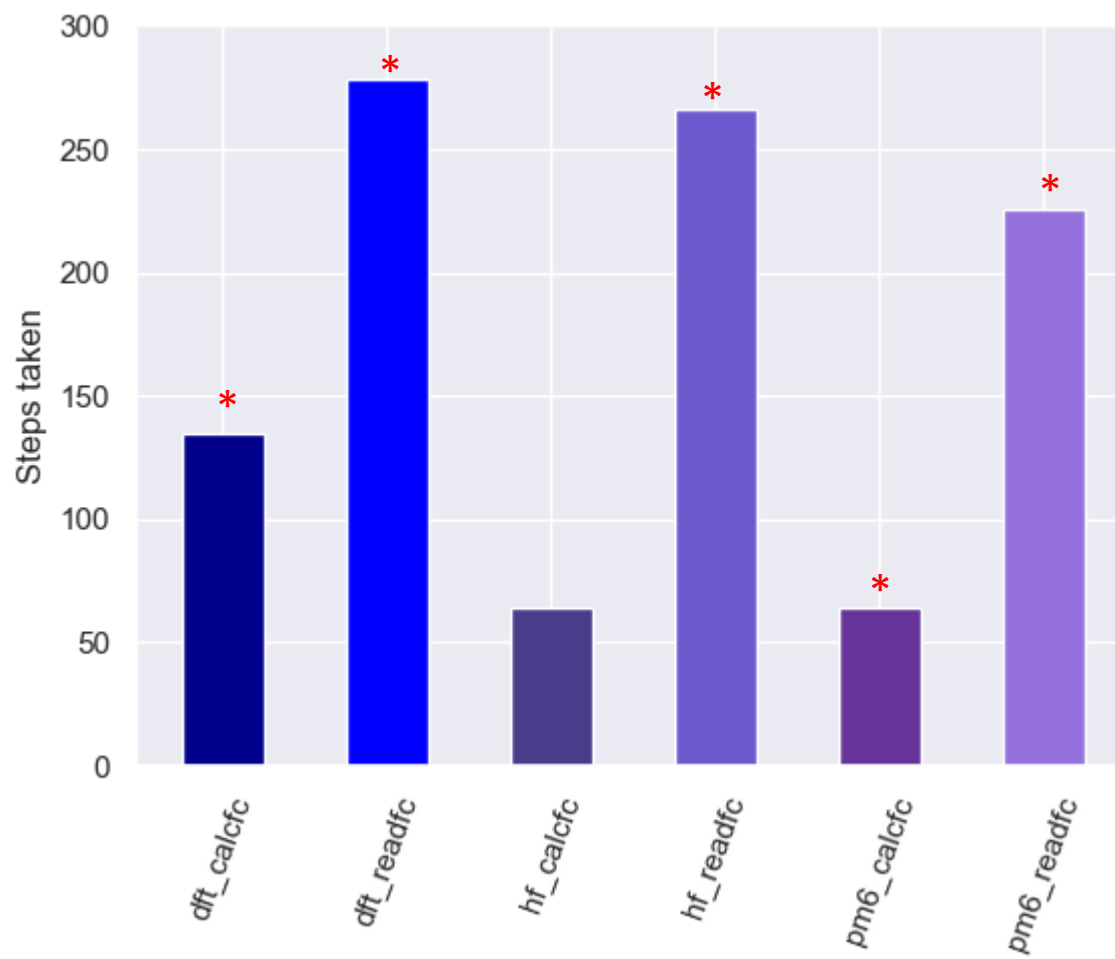
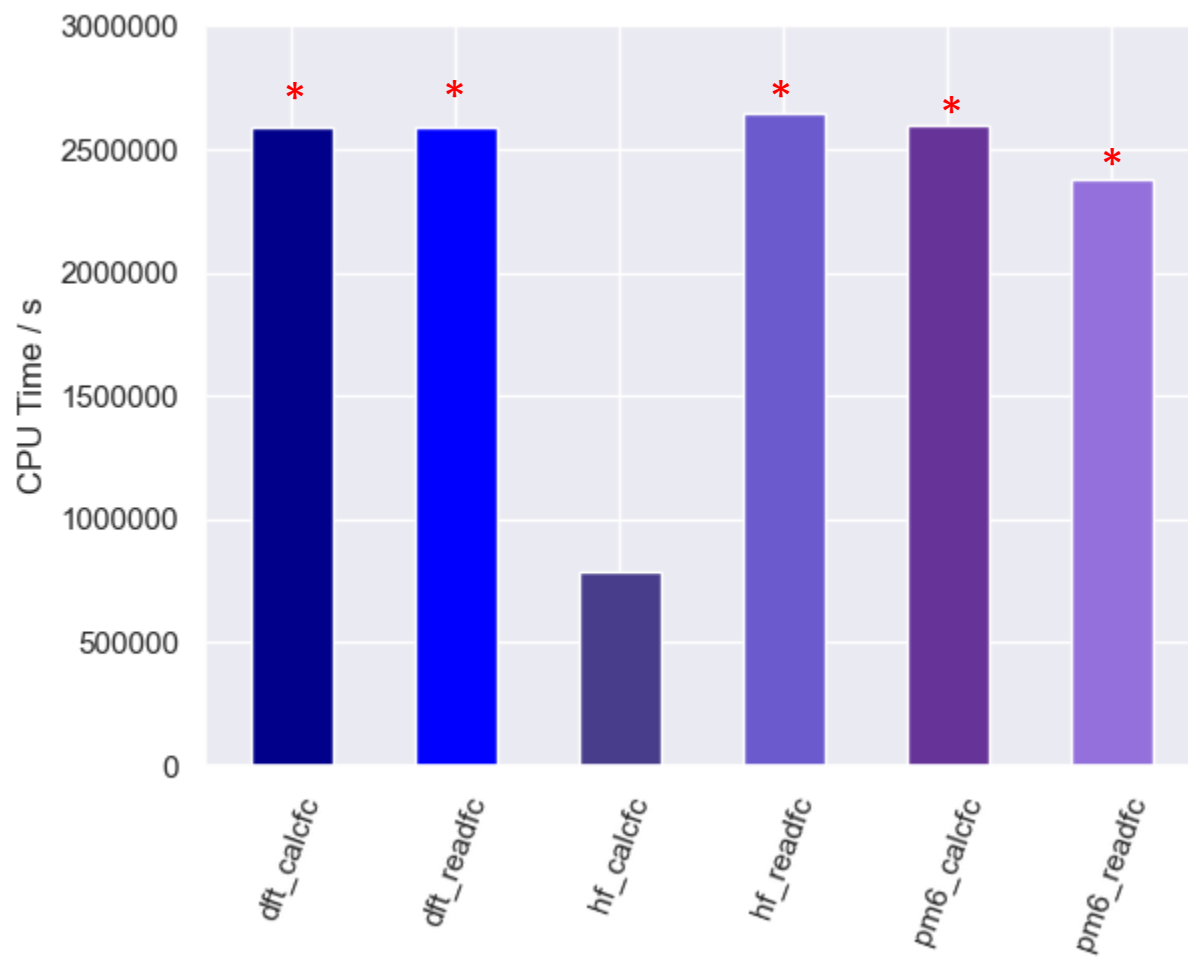
DFT calcfc
(no other jobs converged)



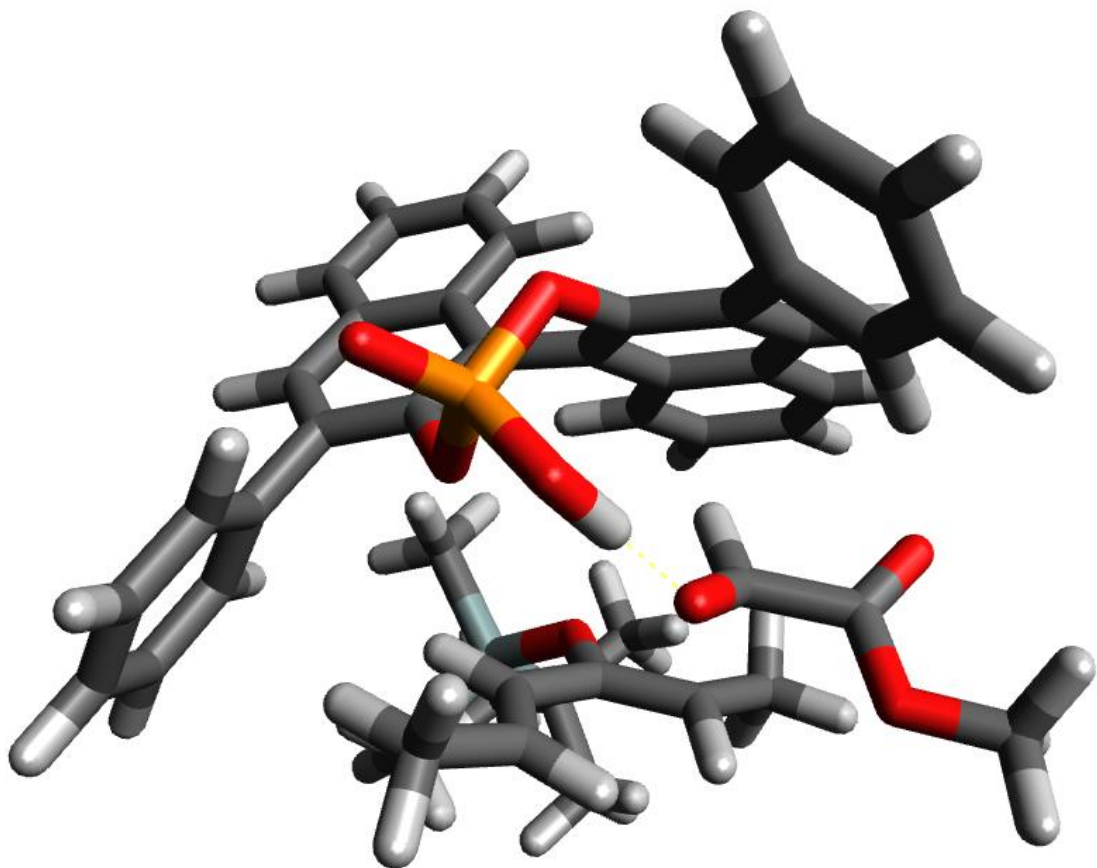
BPA_1



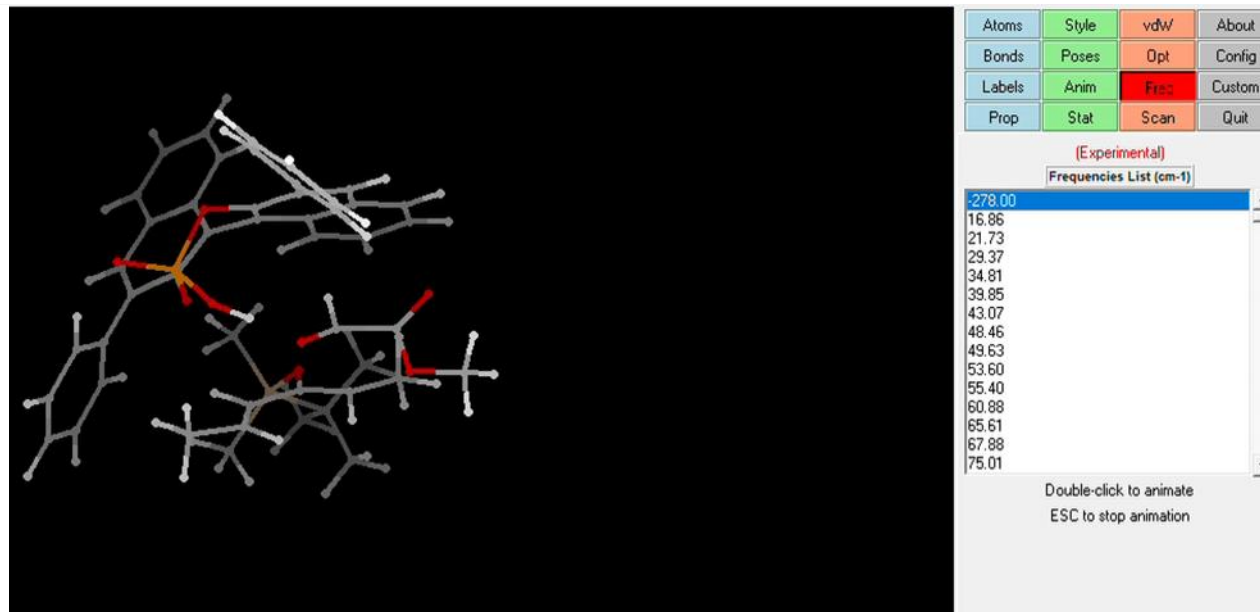
BPA_2



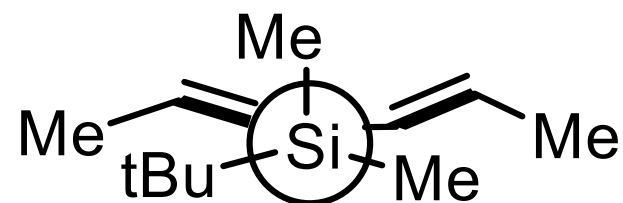
BPA_2: conformation



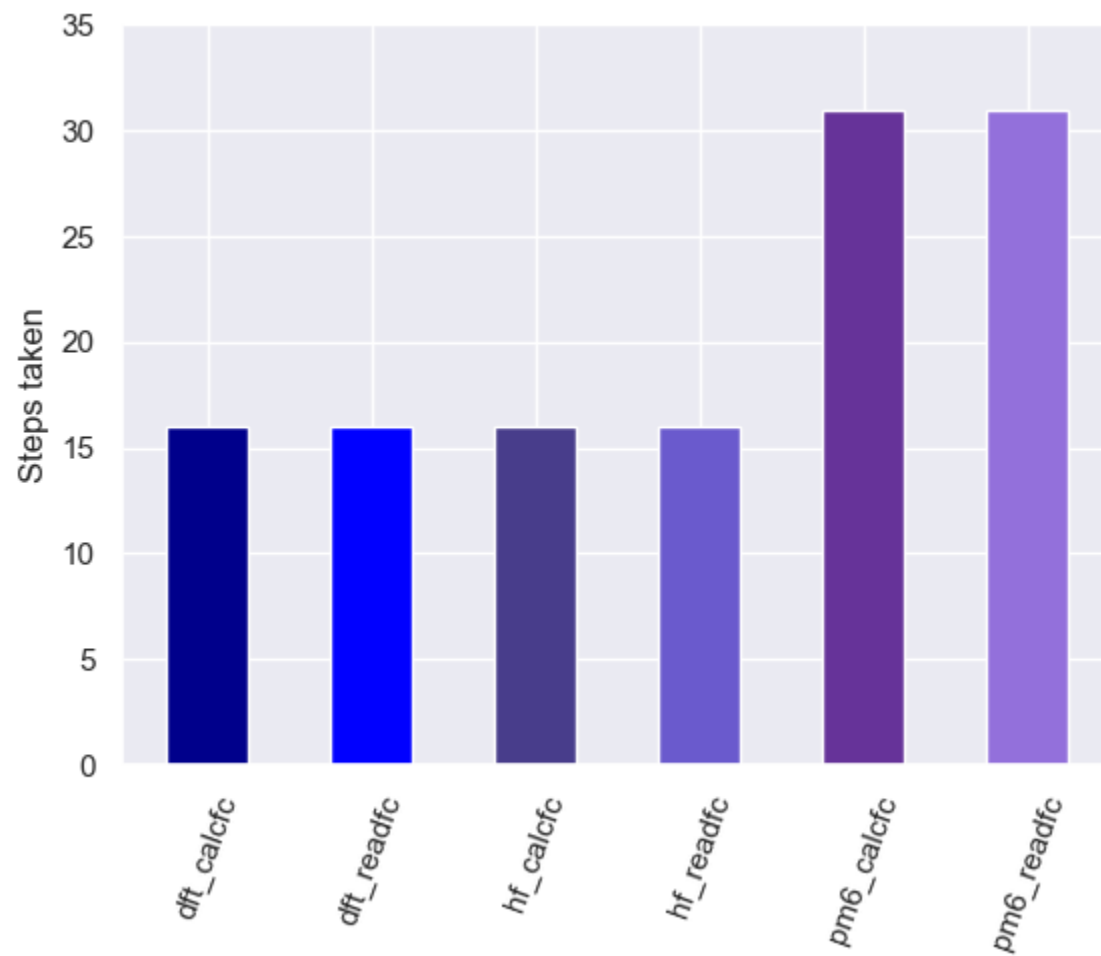
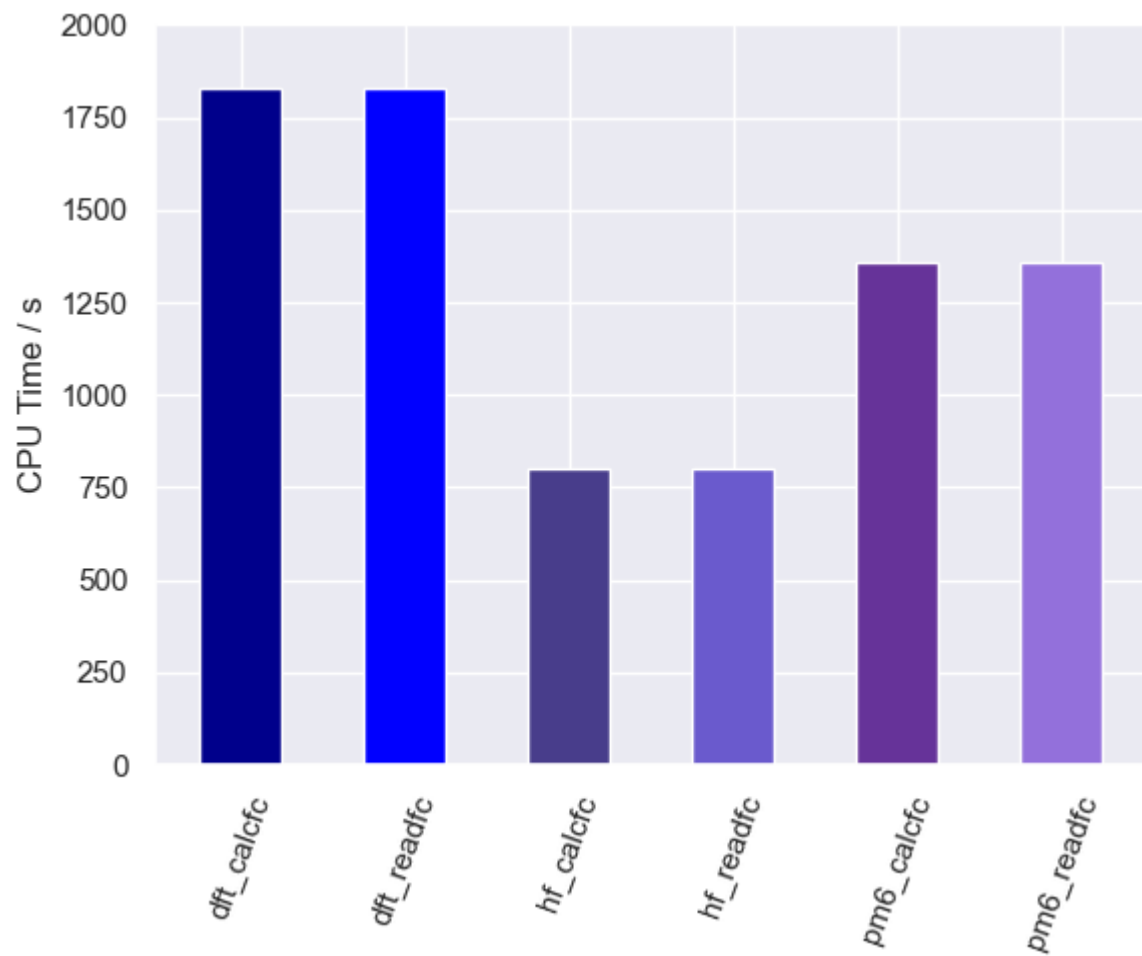
HF calcfc
(no other jobs converged)



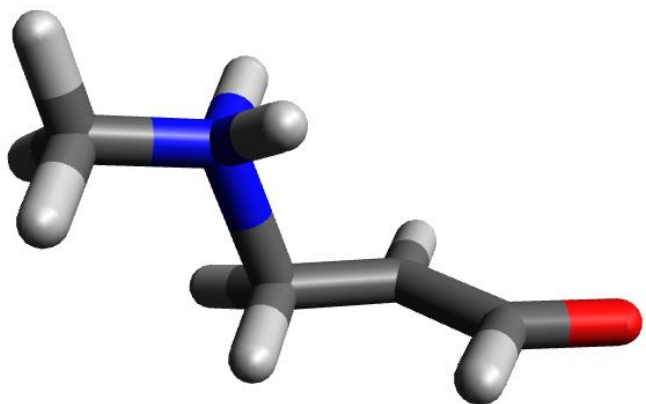
BPA_2



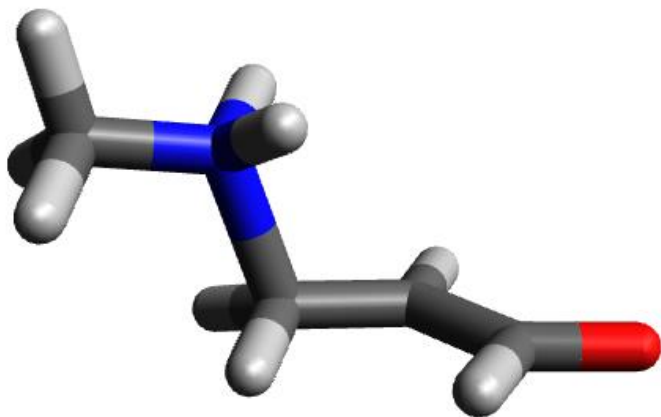
AM_249



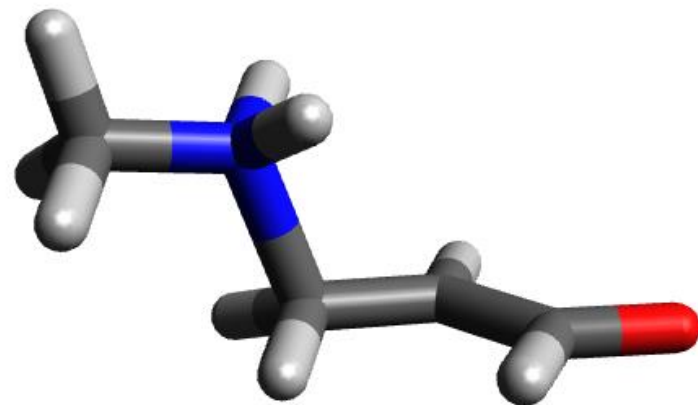
AM_249: conformations



DFT Hessian

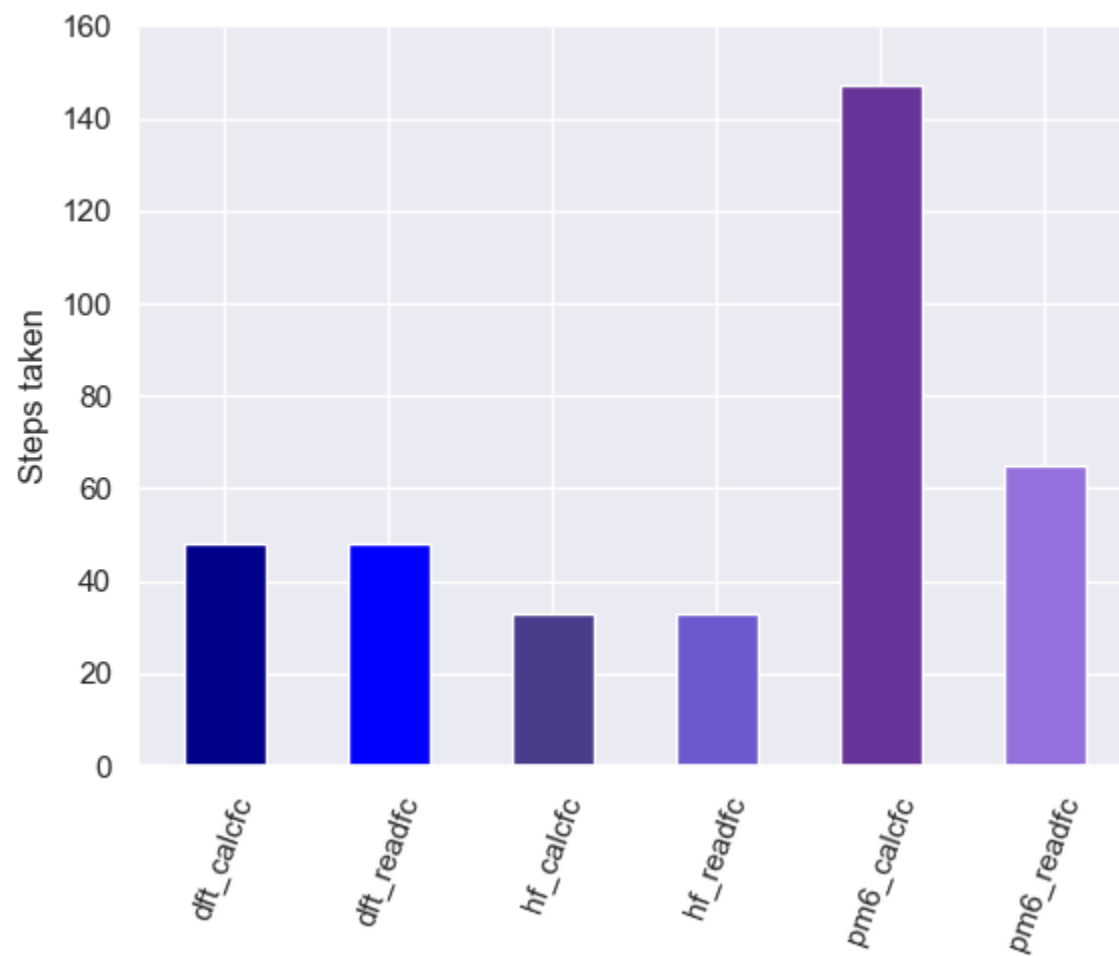
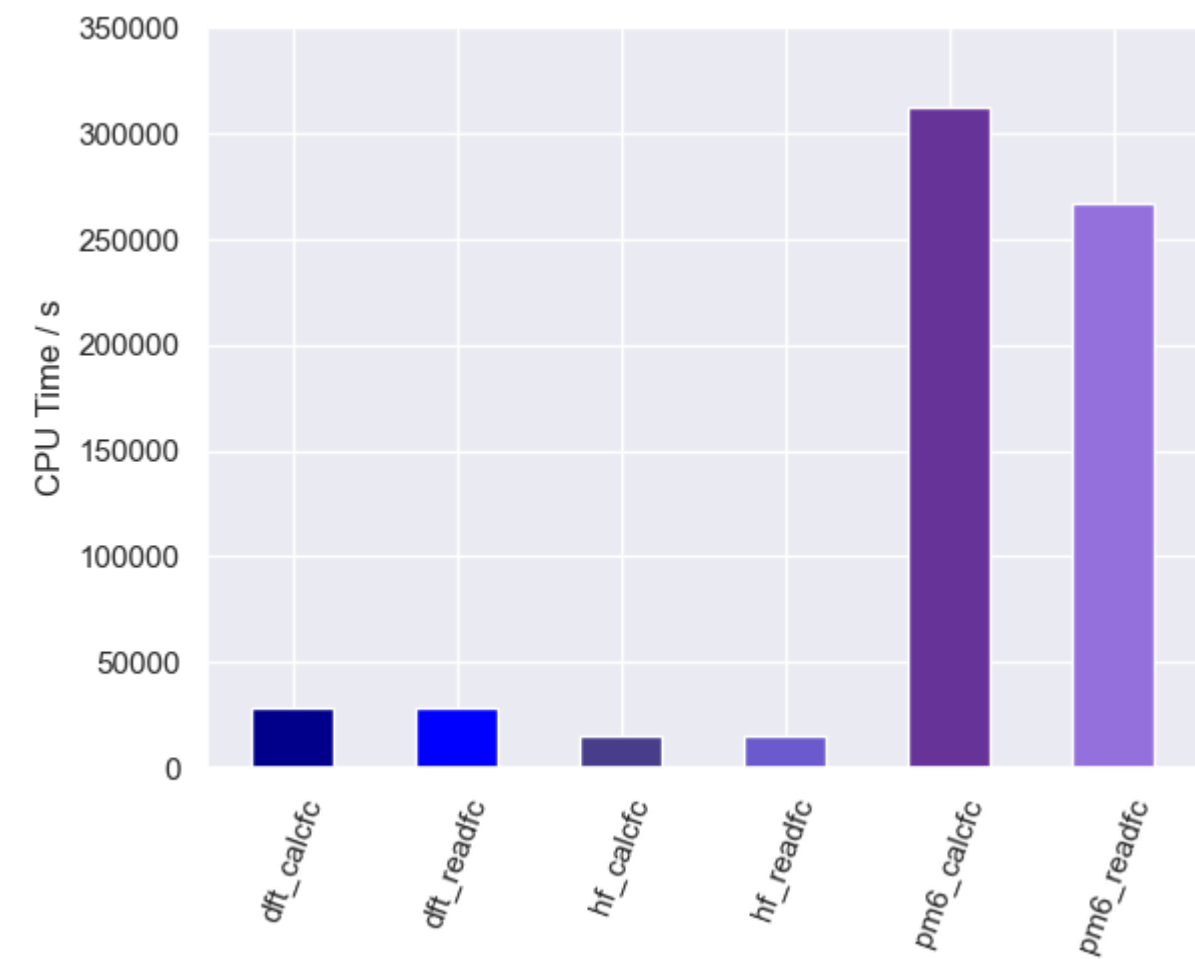


HF Hessian

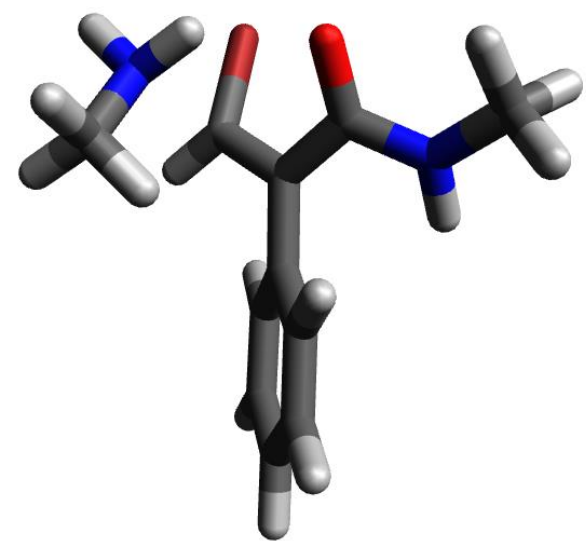


SQM Hessian

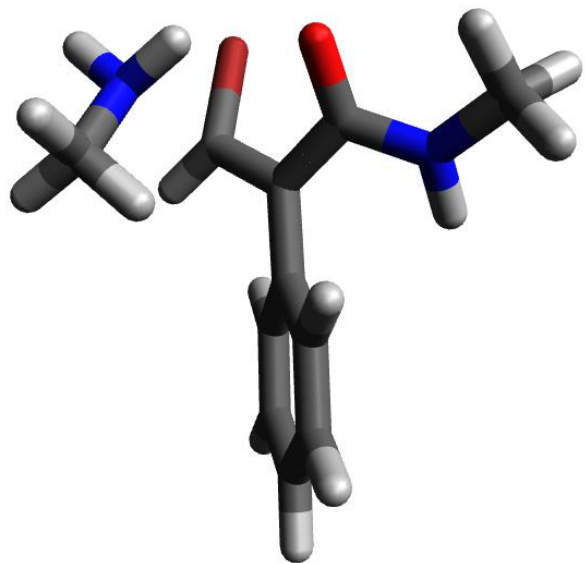
AM_325



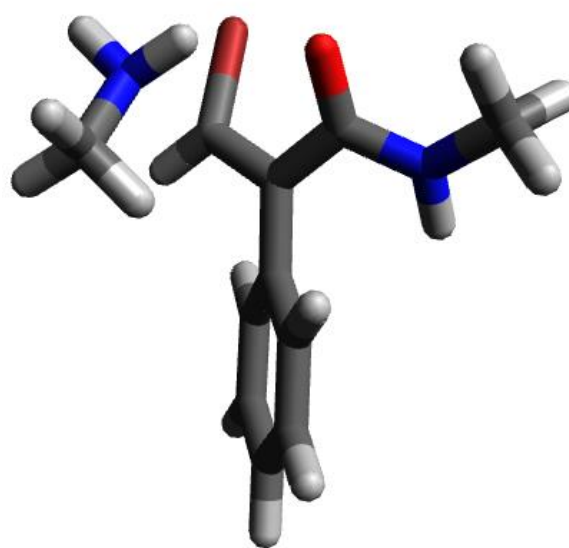
AM_325: conformations



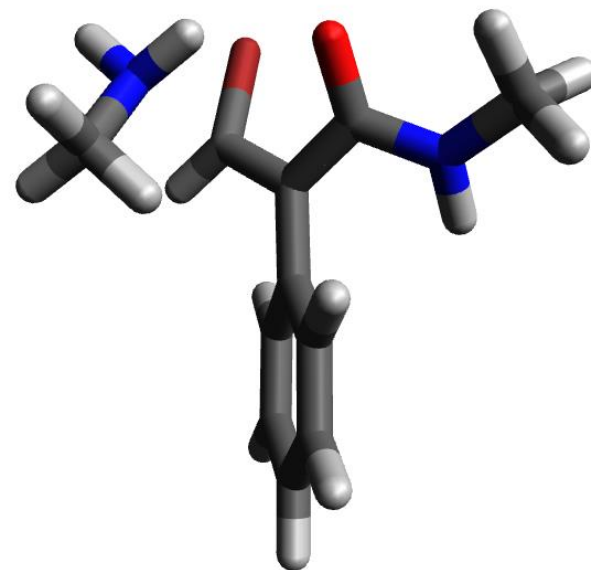
DFT Hessian



HF Hessian

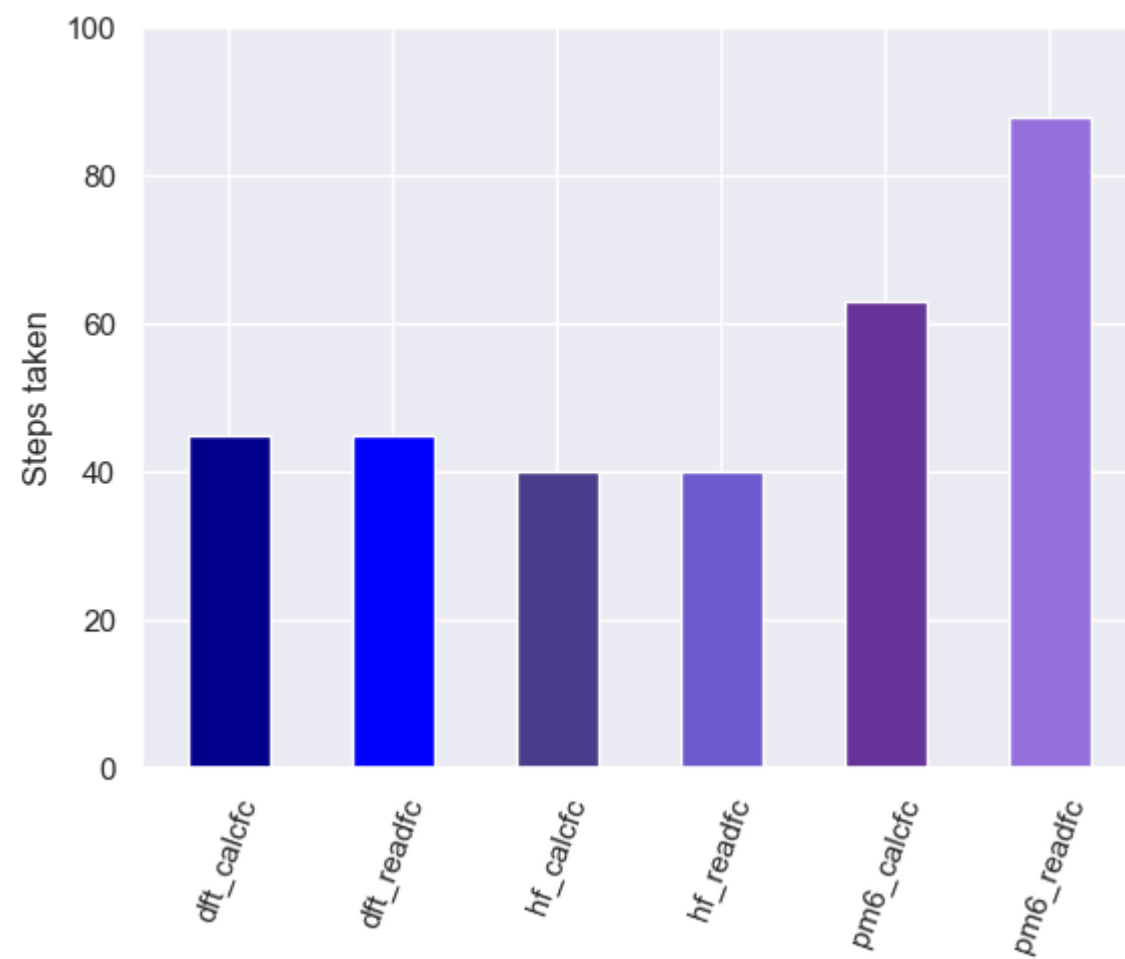
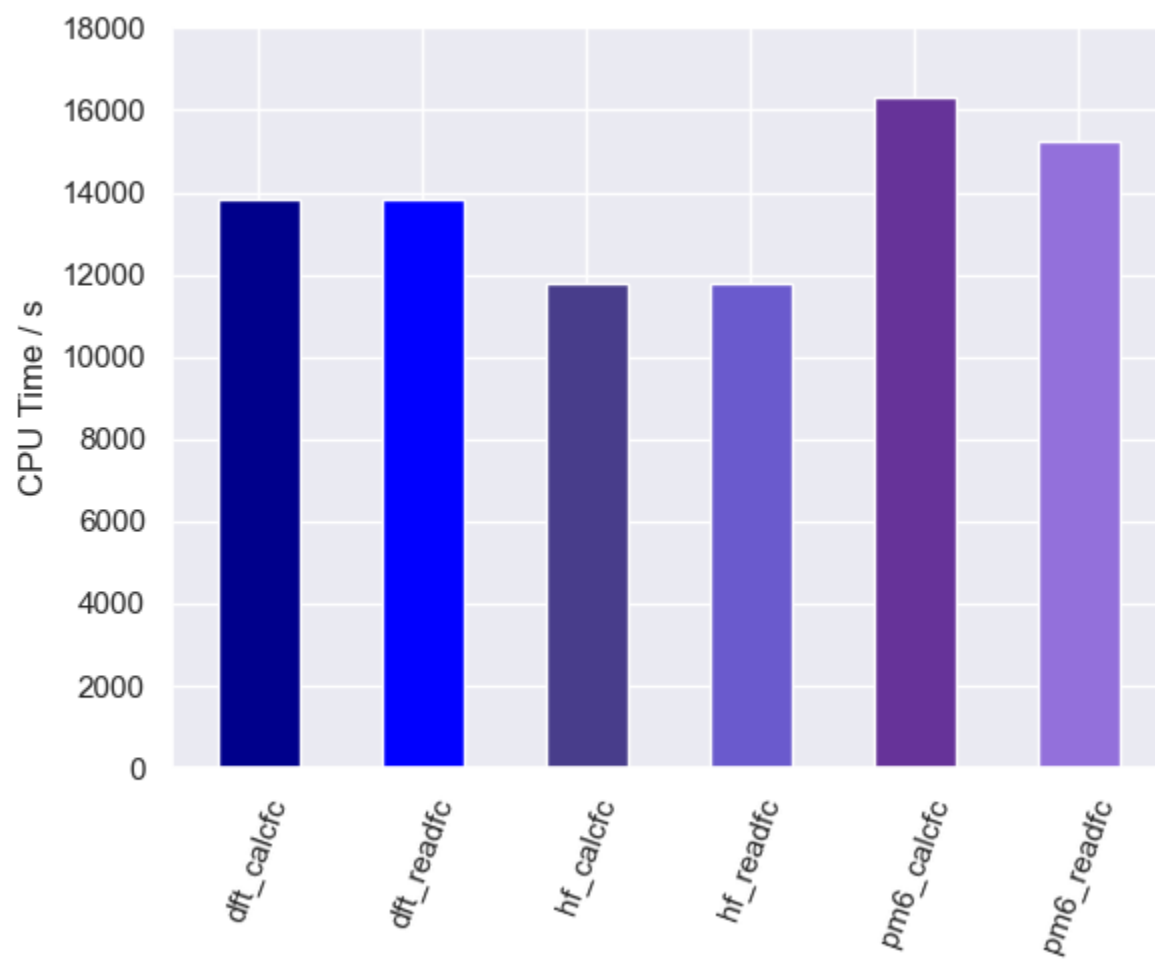


SQM_calcfc

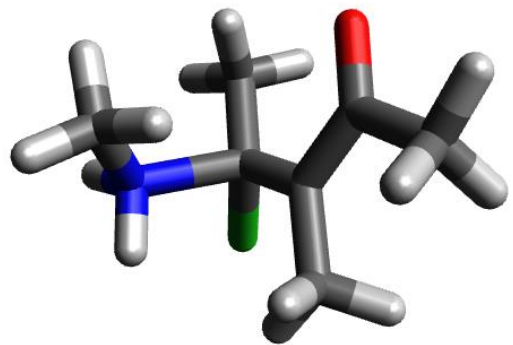


SQM_readfc

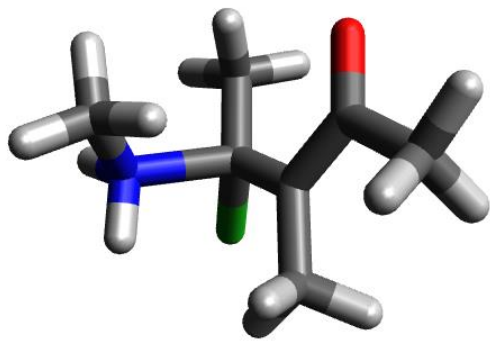
AM_898



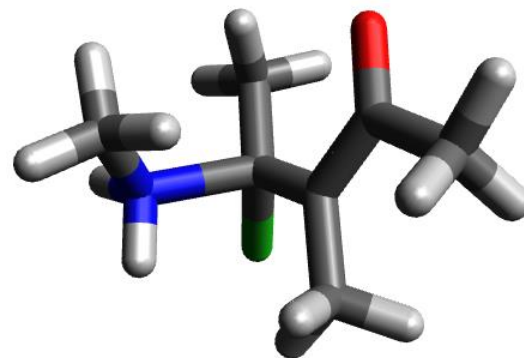
AM_898: conformations



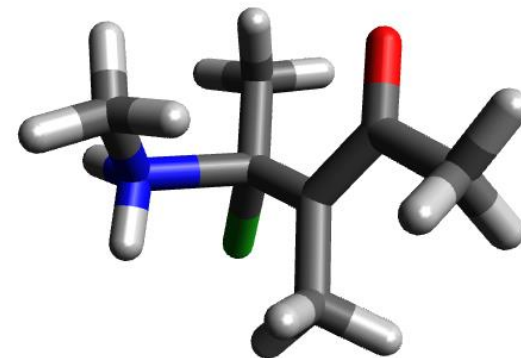
DFT Hessian



HF Hessian

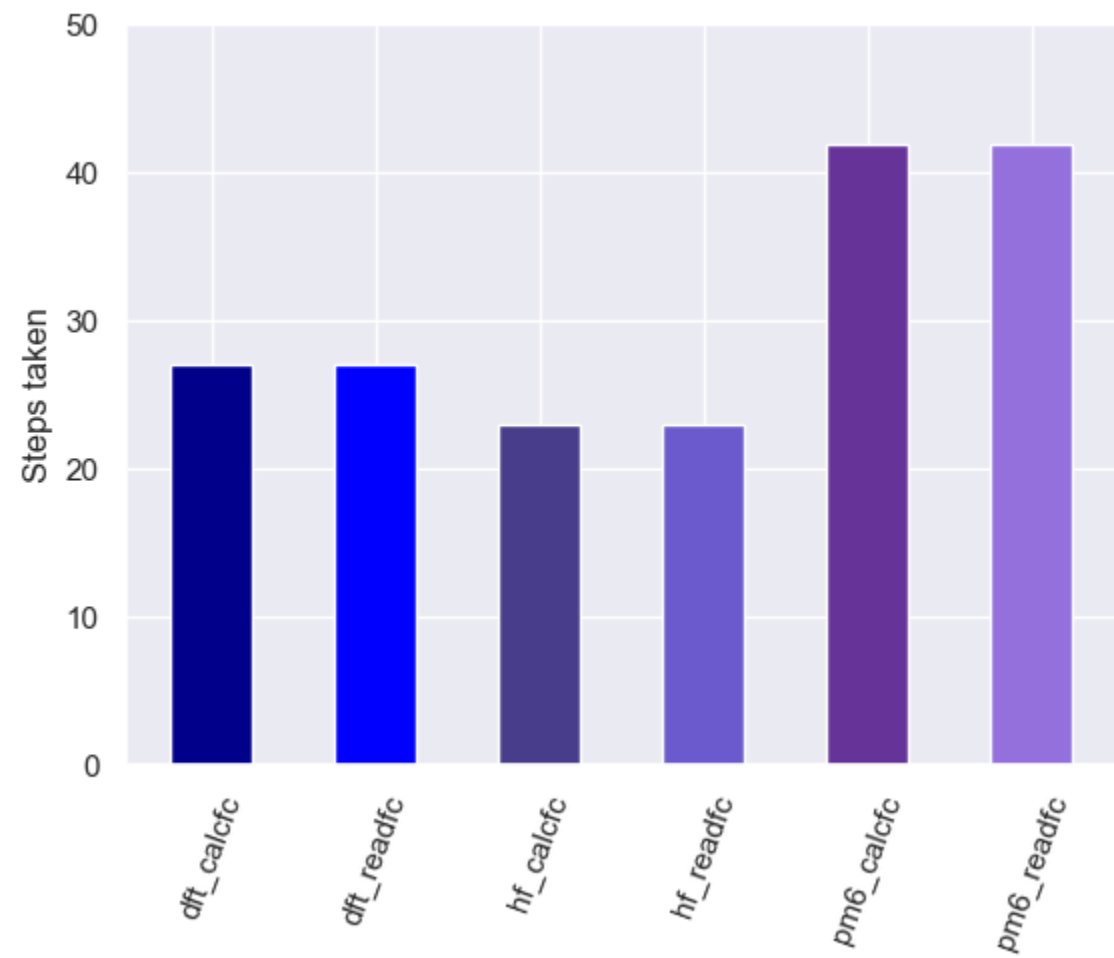
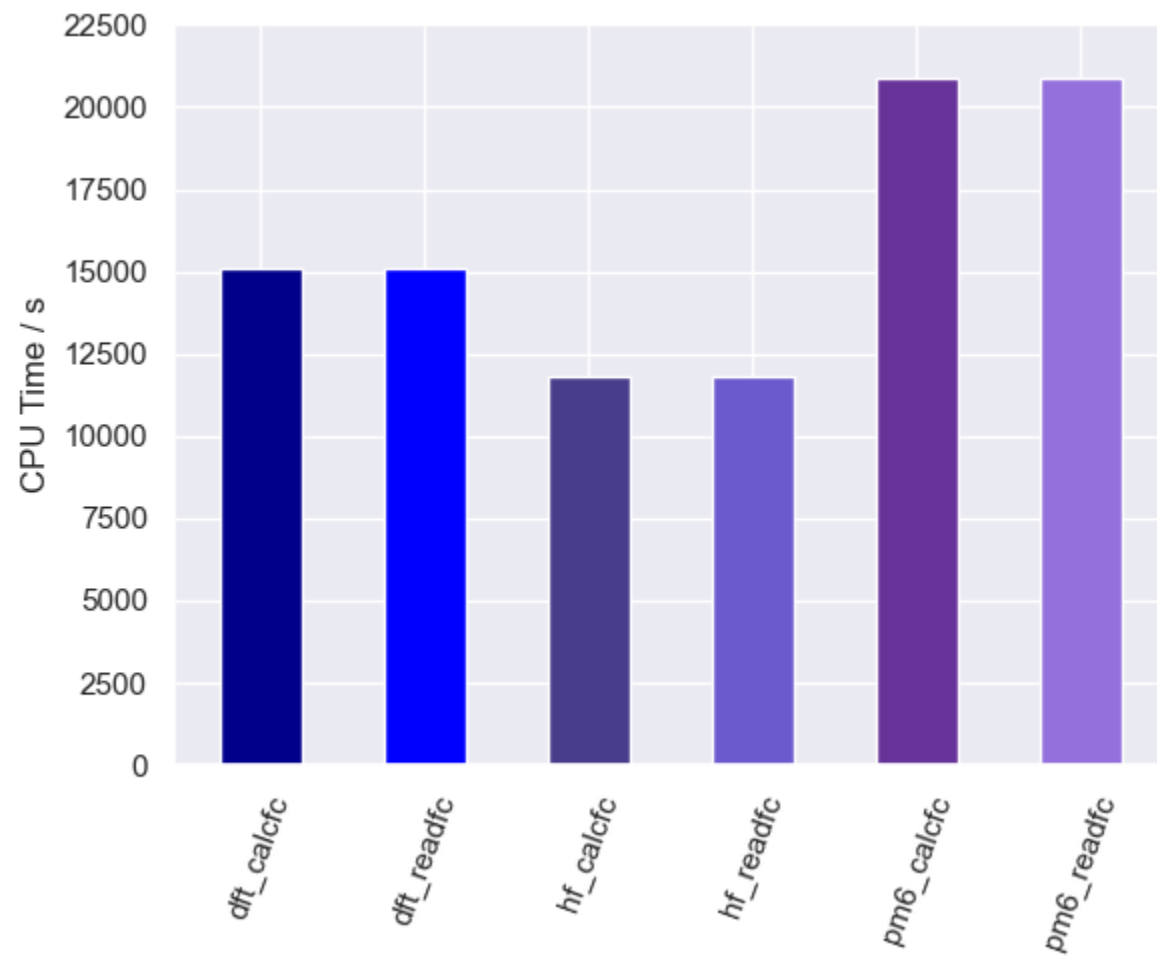


SQM_calcfc

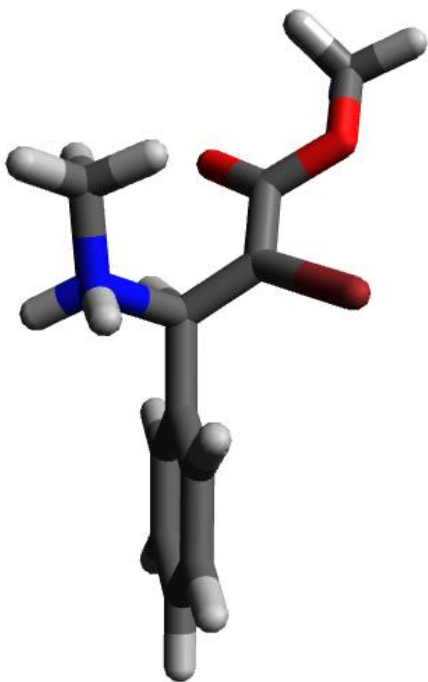


SQM_readfc

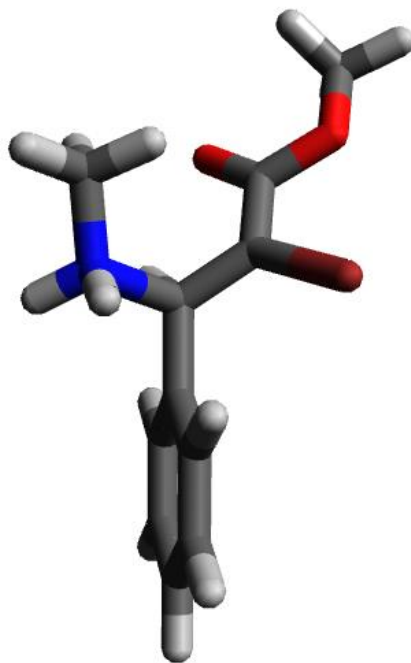
AM_68



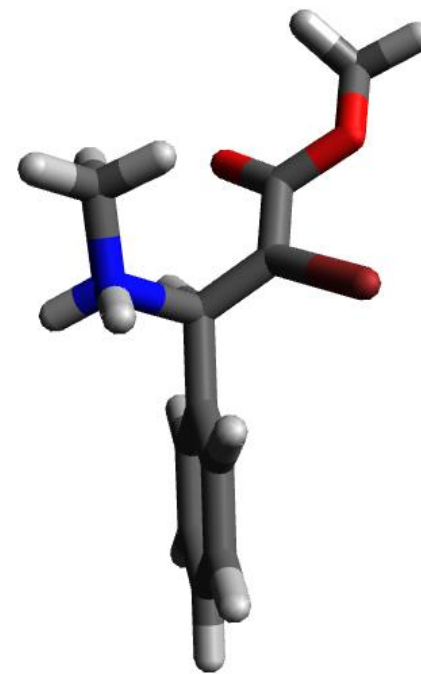
AM_68: conformations



DFT Hessian

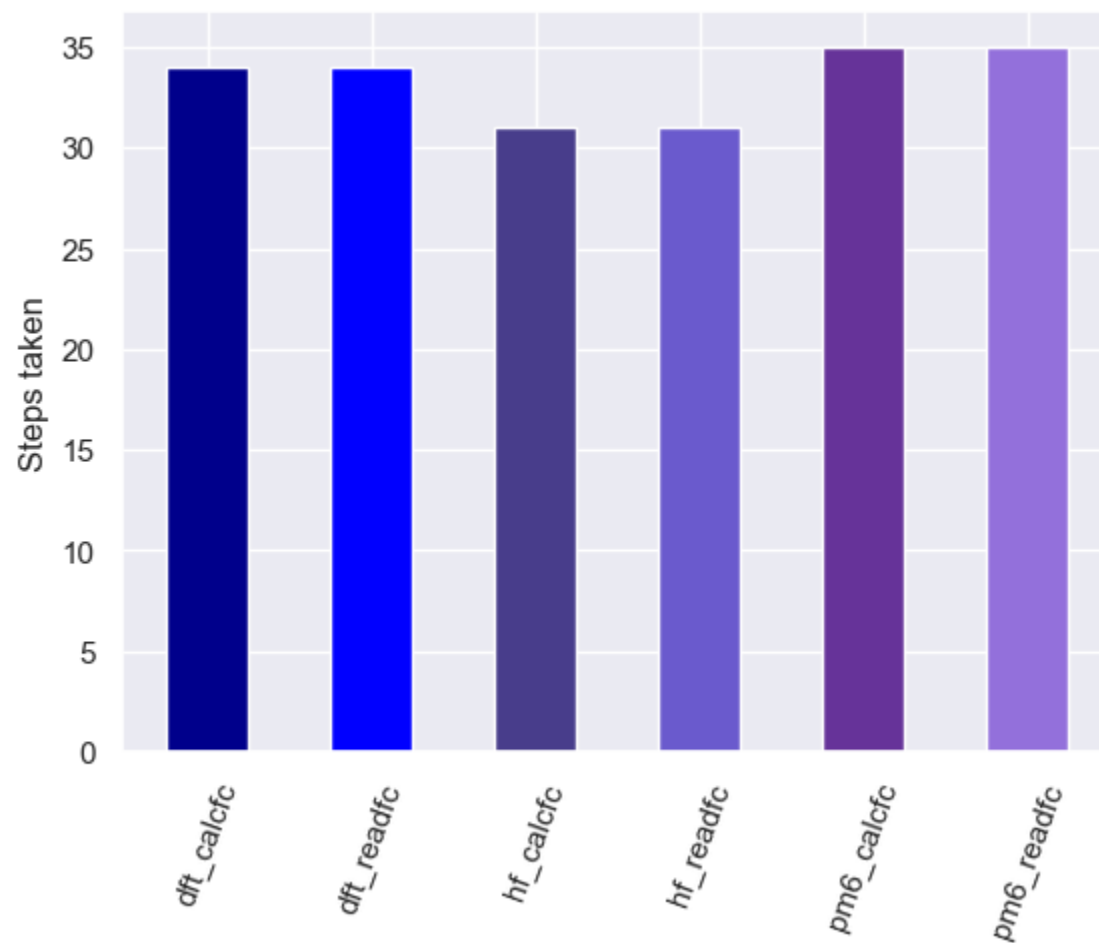
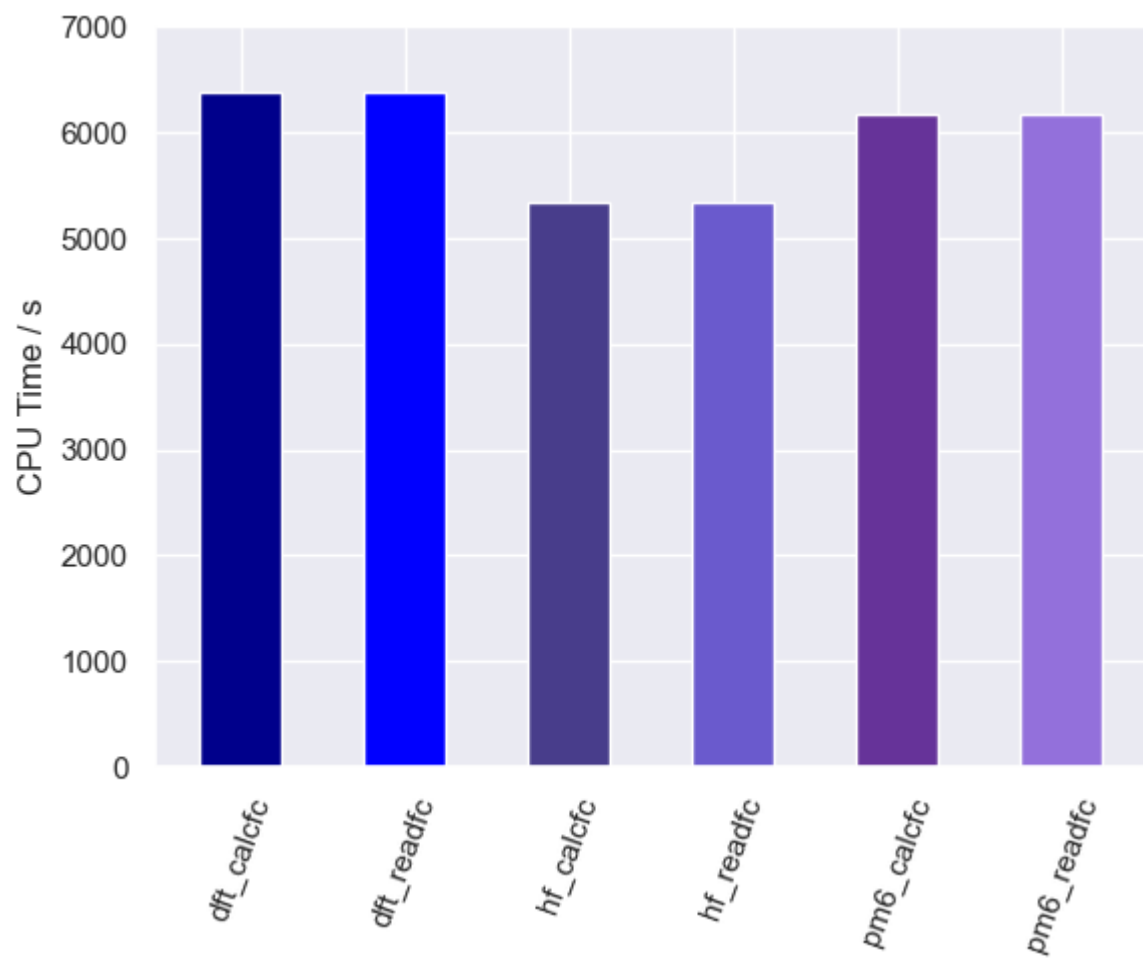


HF Hessian

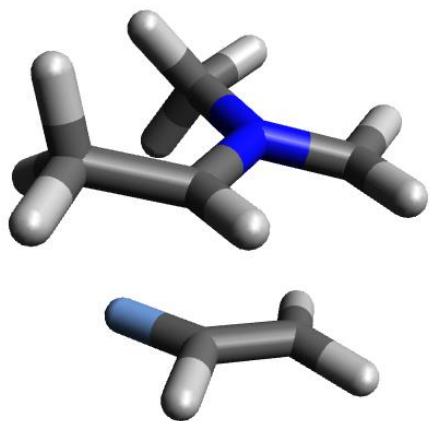


SQM Hessian

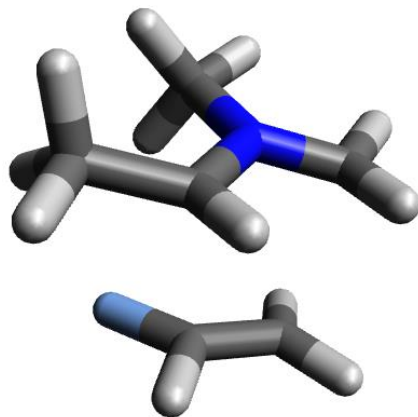
allylic



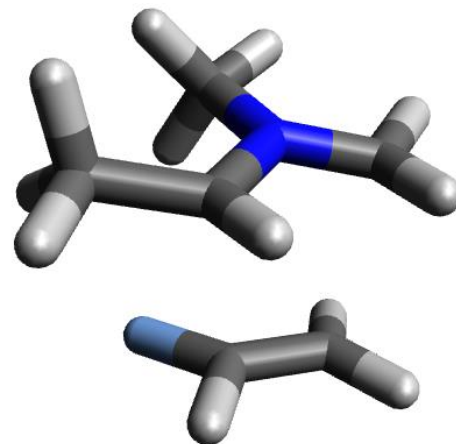
allylic: conformations



DFT Hessian

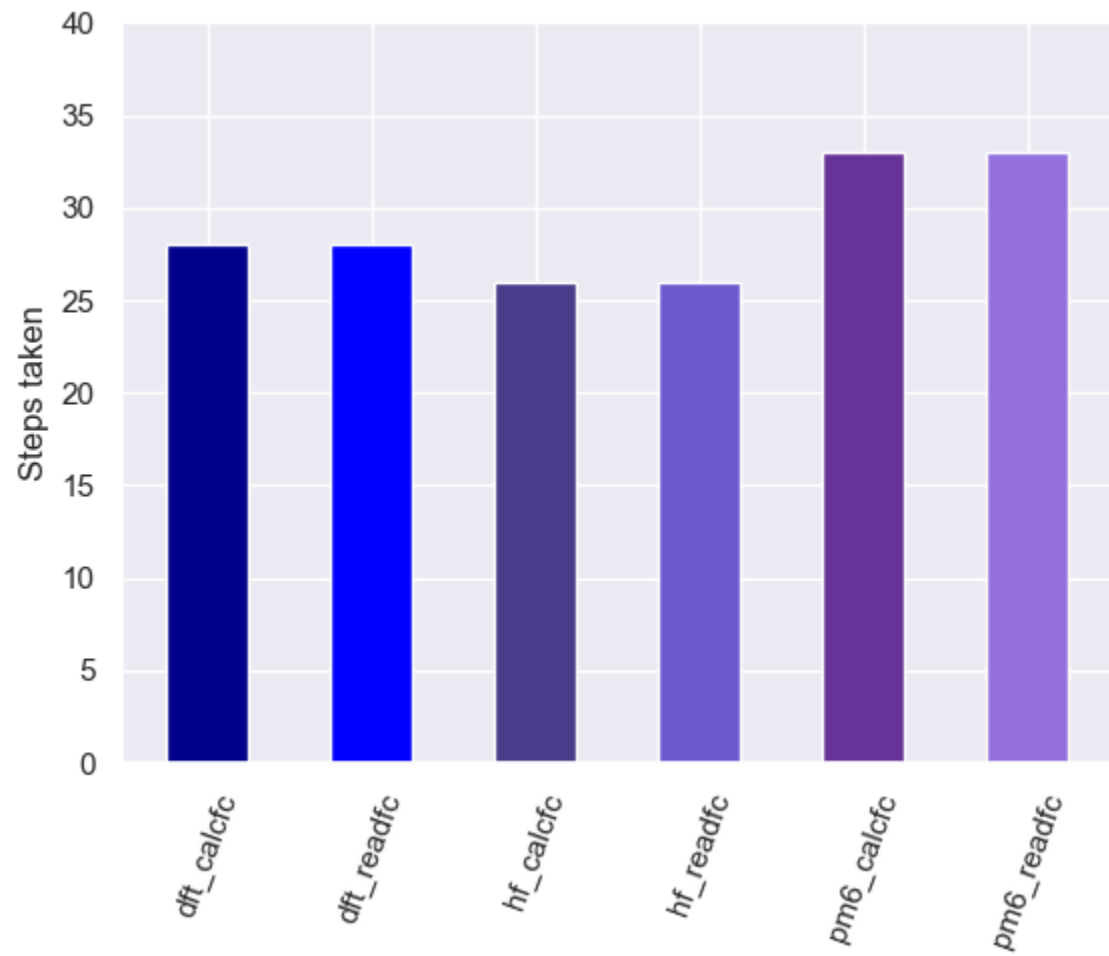
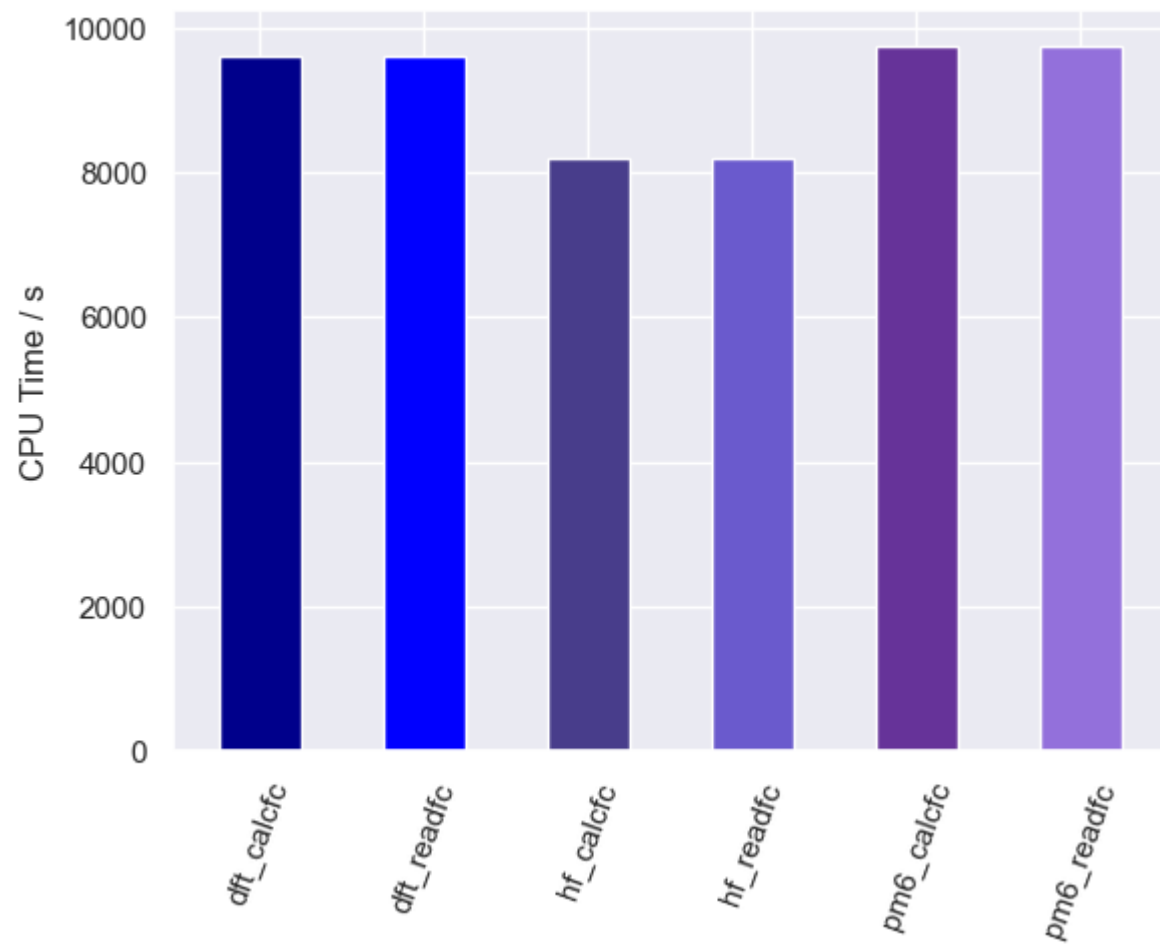


HF Hessian

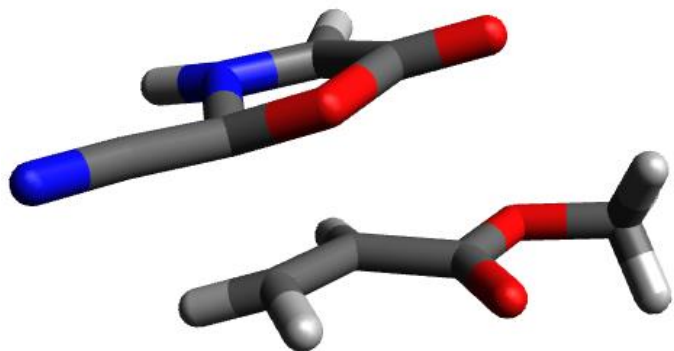


SQM Hessian

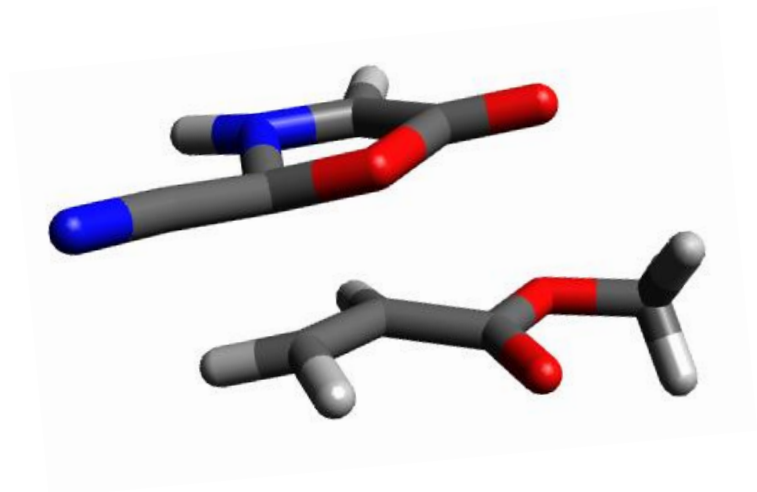
cyclic



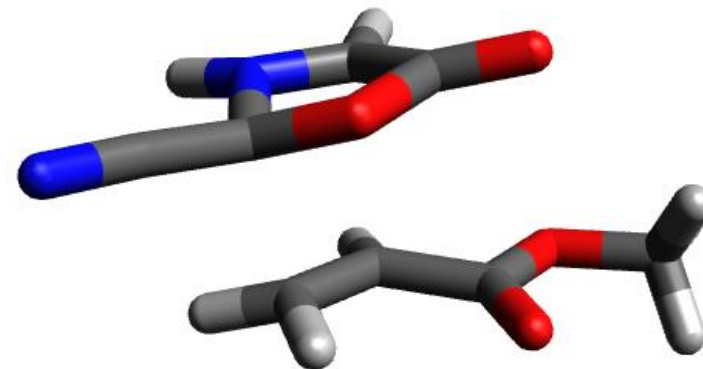
cyclic: conformations



DFT Hessian

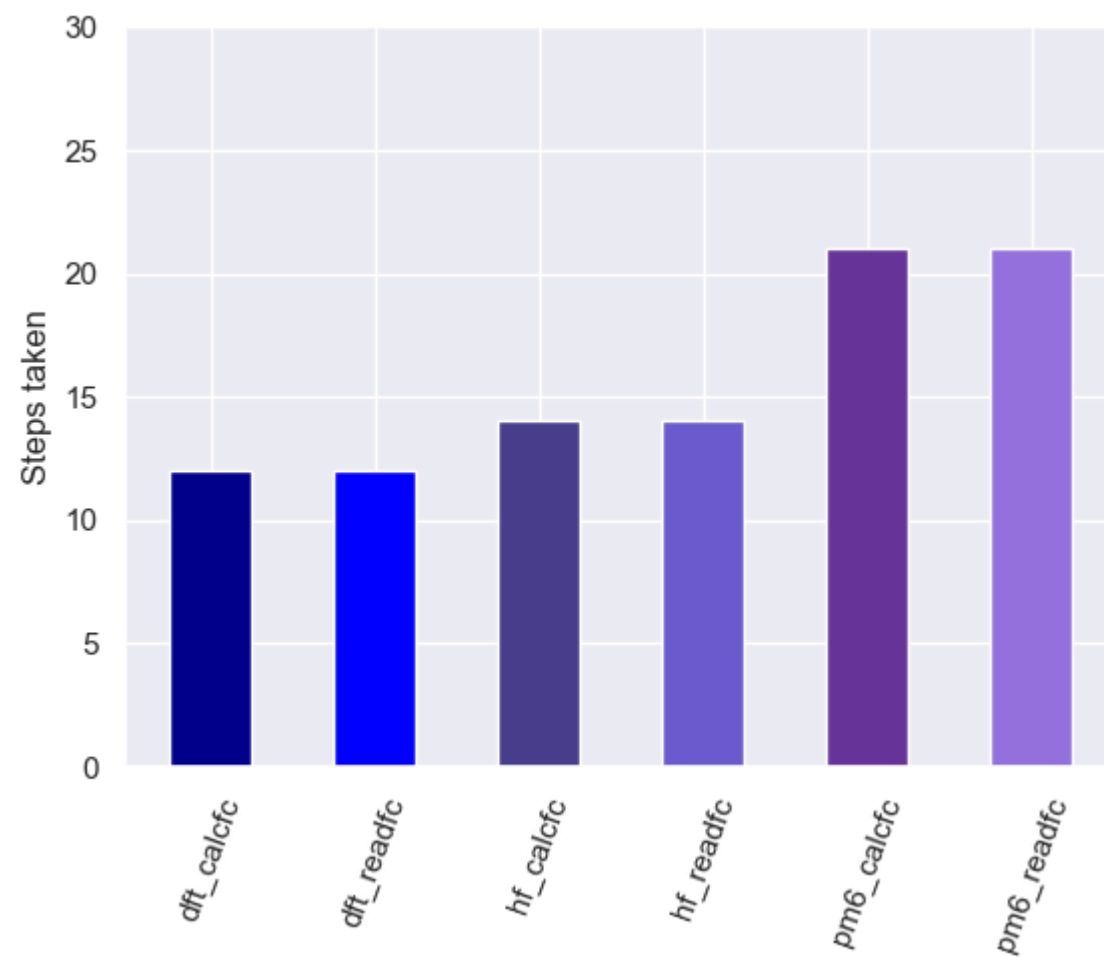
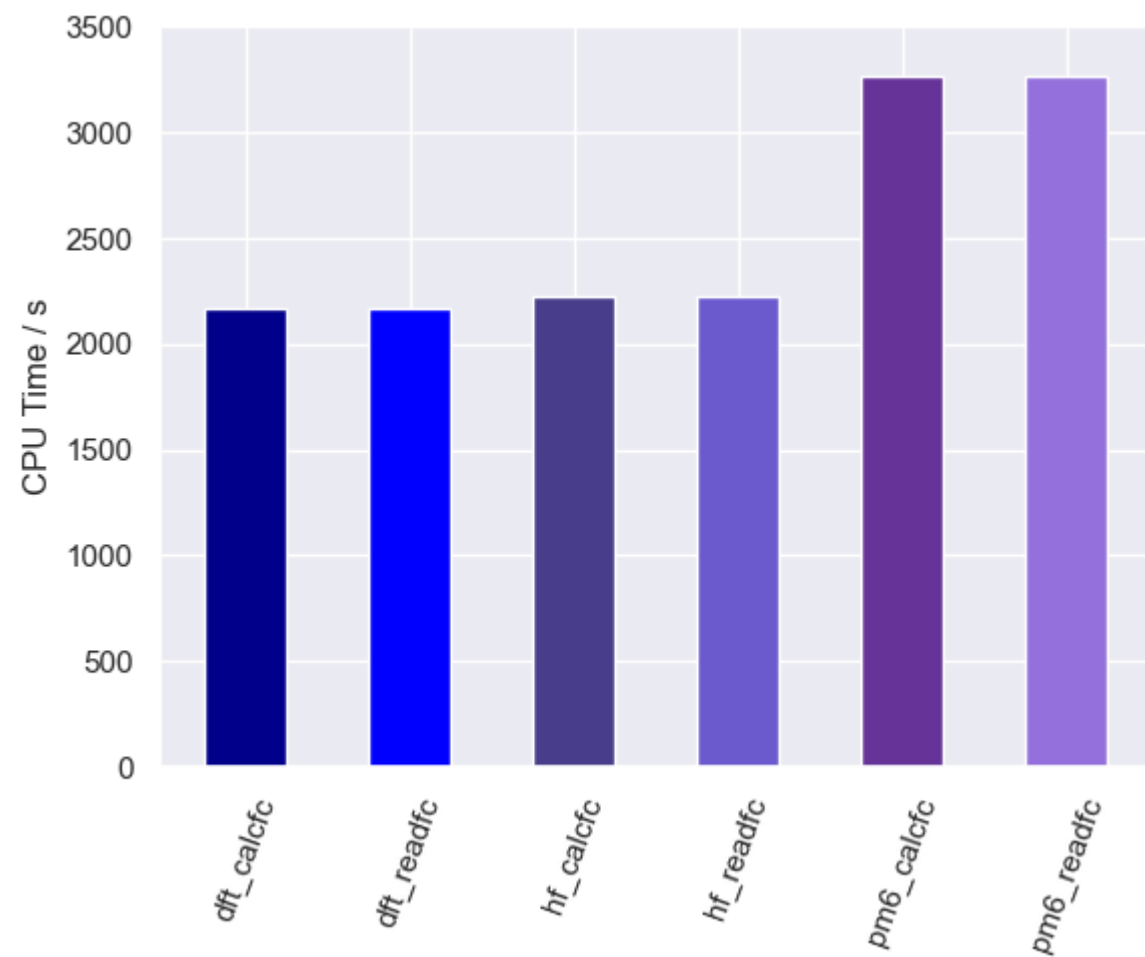


HF Hessian

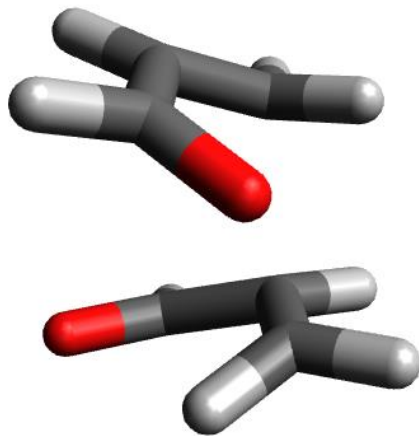


SQM Hessian

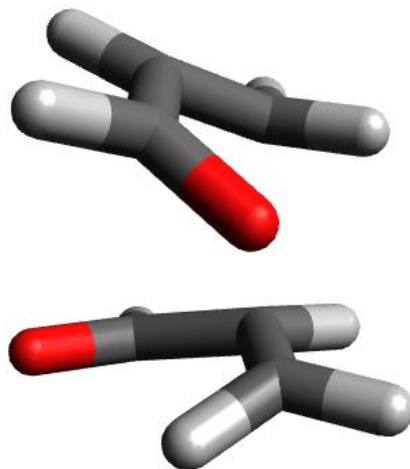
DA_1



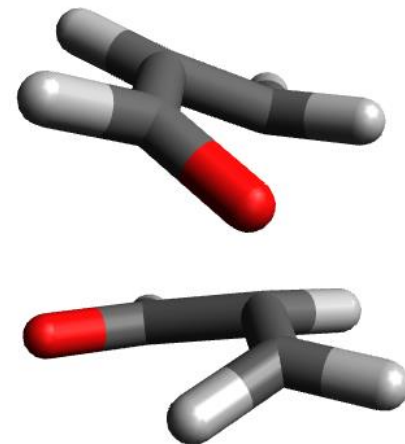
DA_1: conformations



DFT Hessian

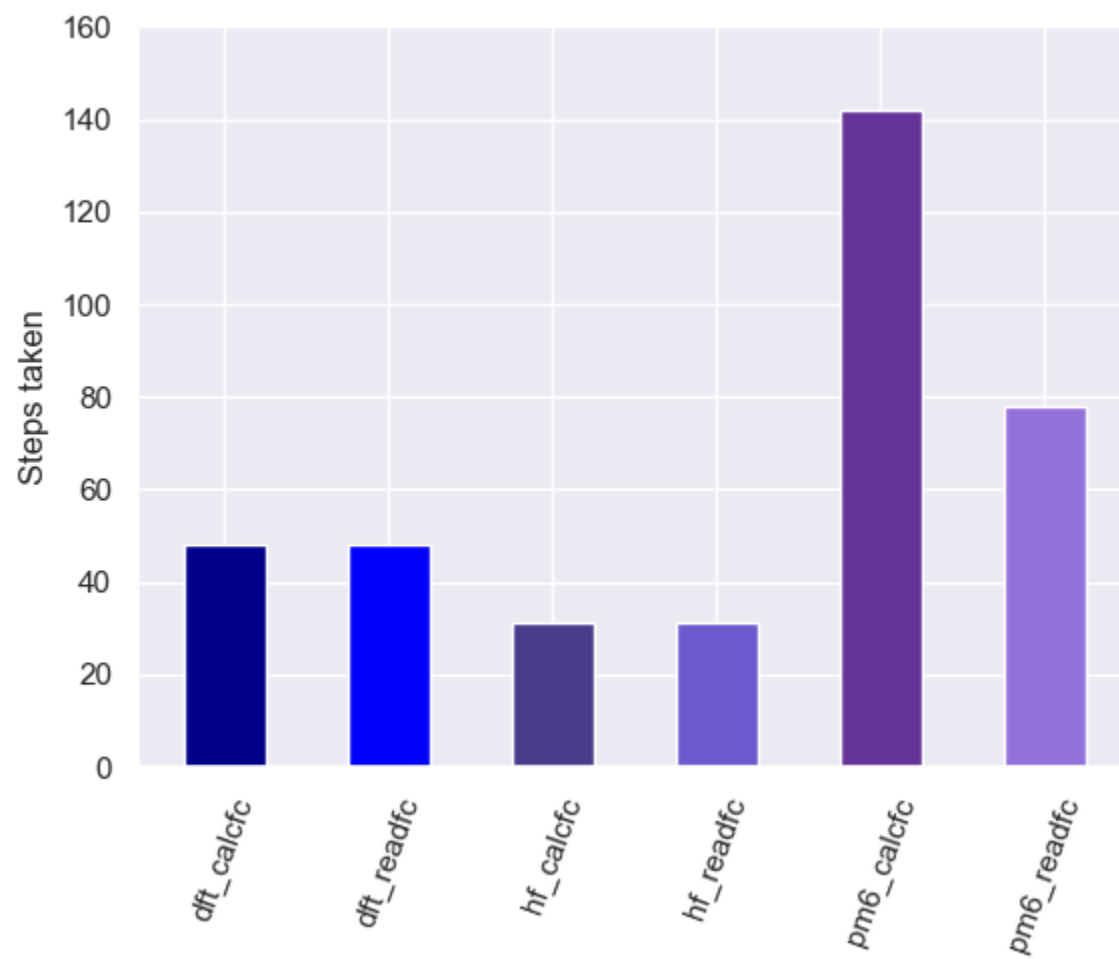
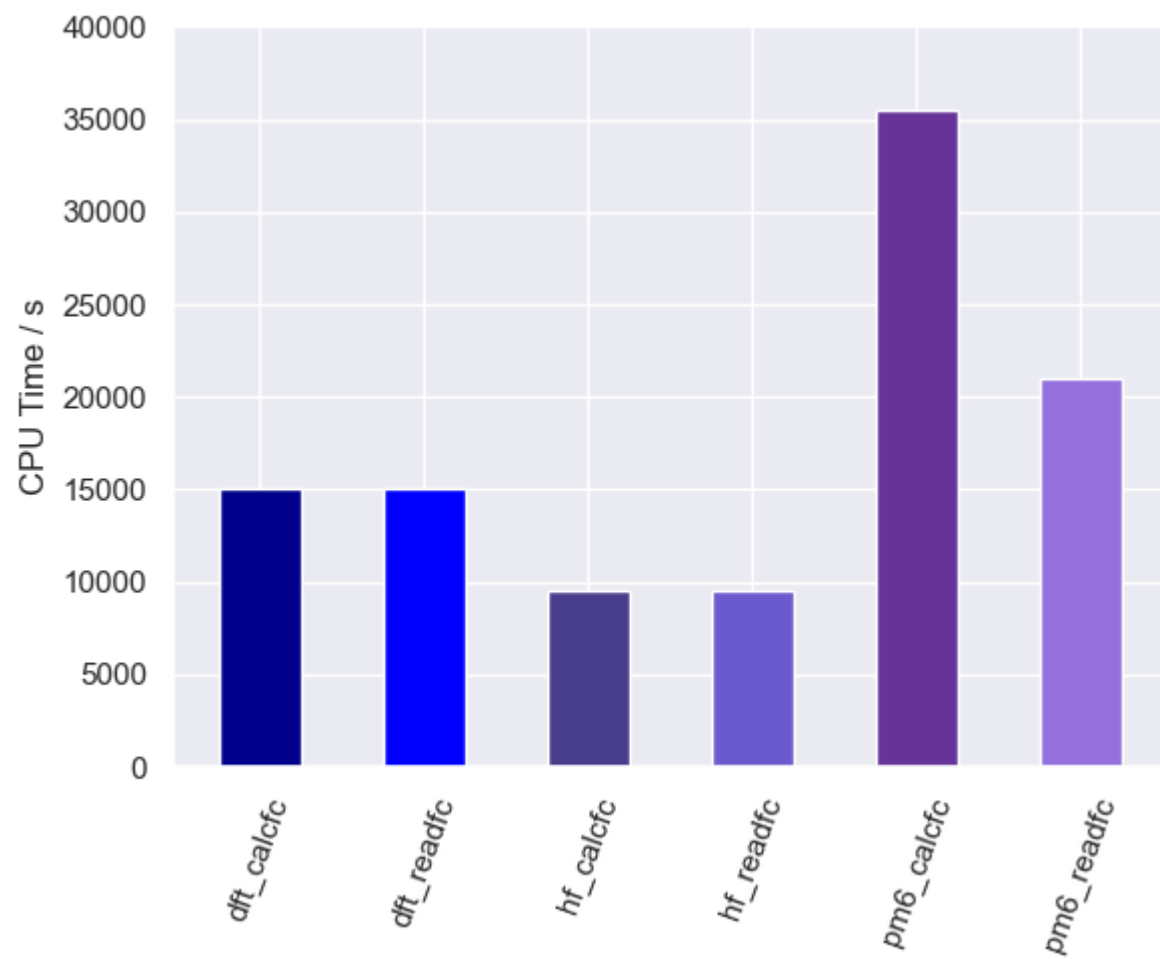


HF Hessian

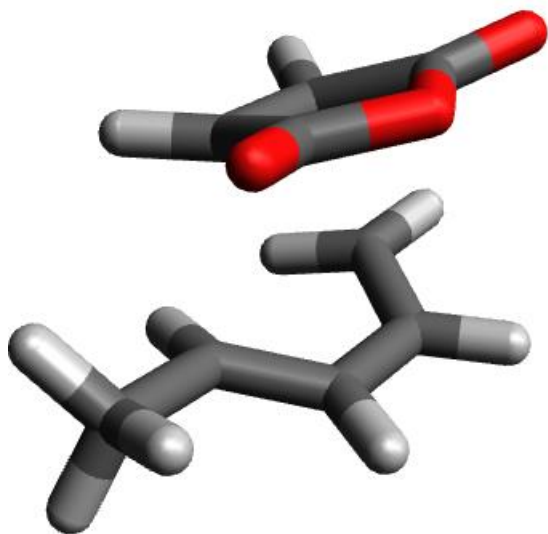


SQM Hessian

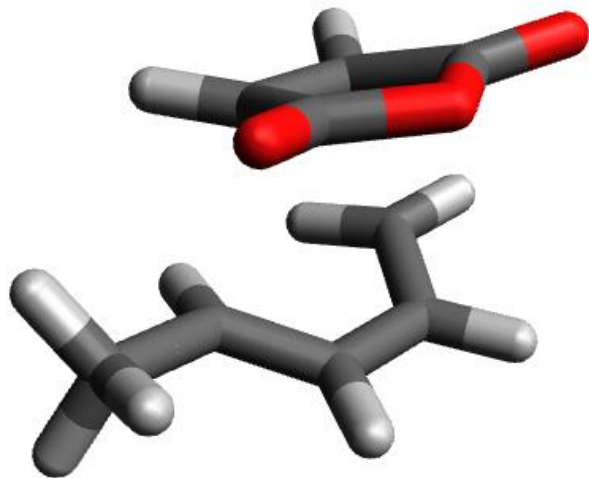
DA_2



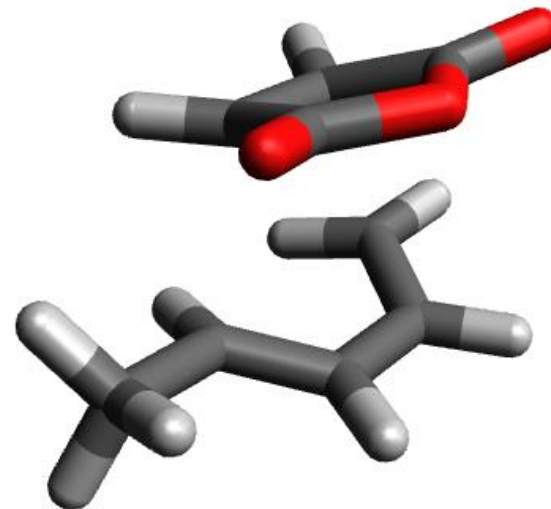
DA_2: conformations



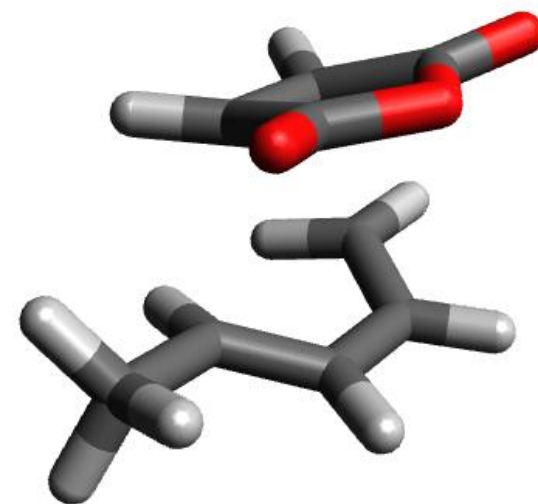
DFT Hessian



HF Hessian

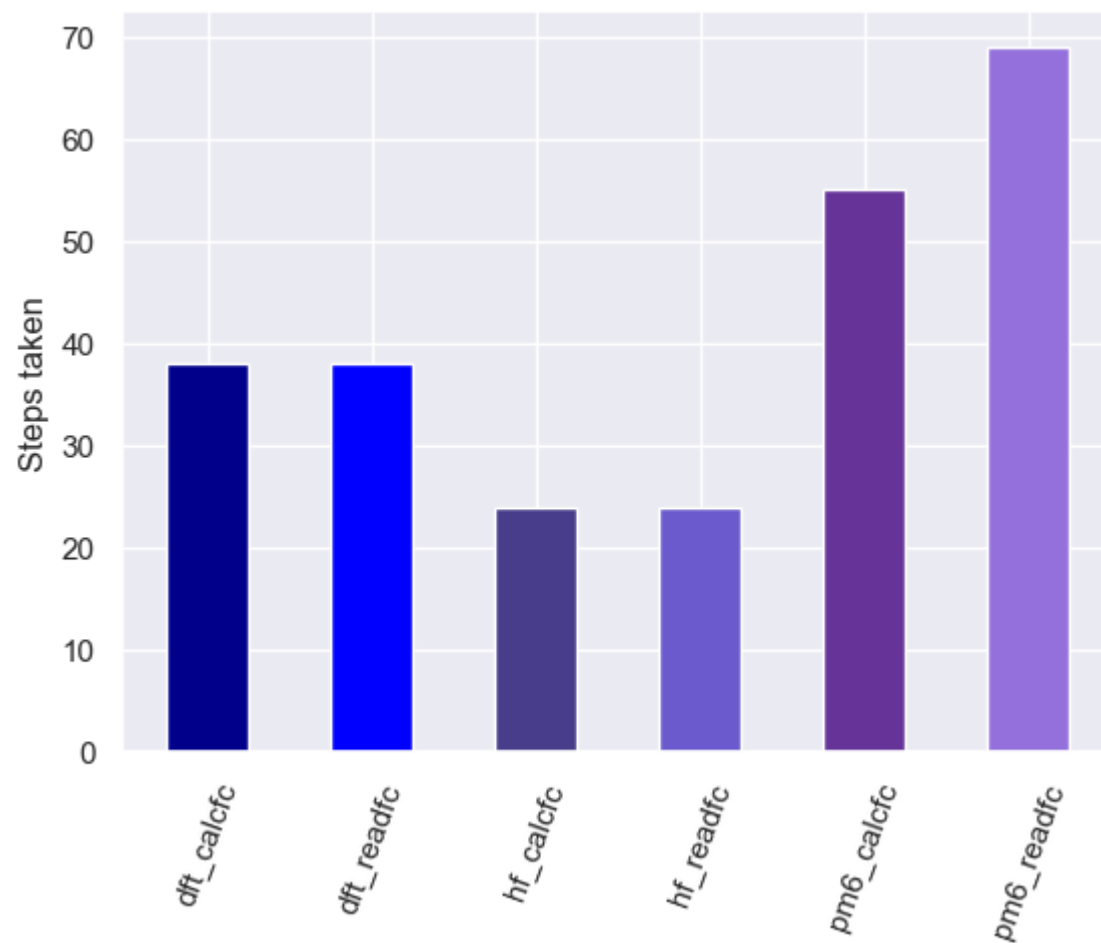
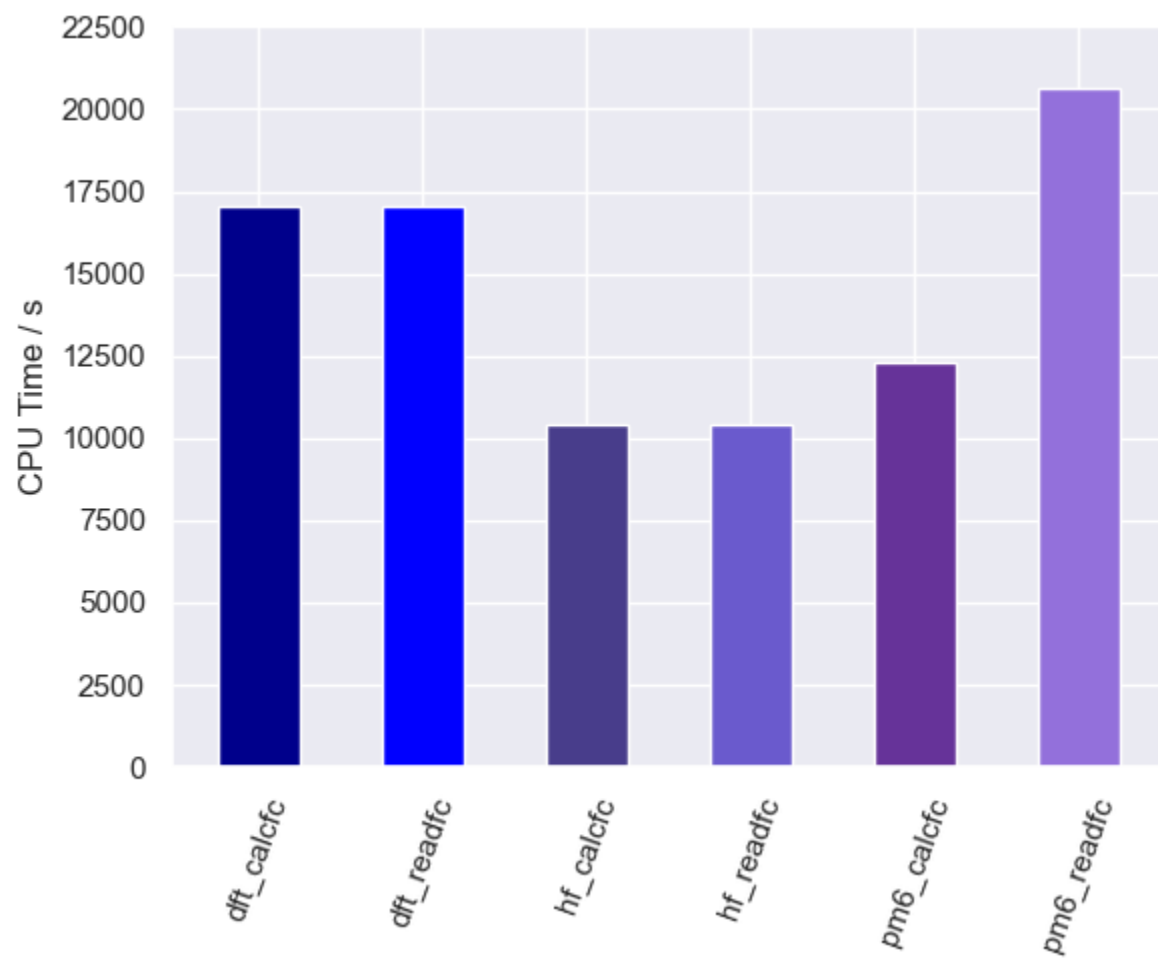


SQM_calcfc

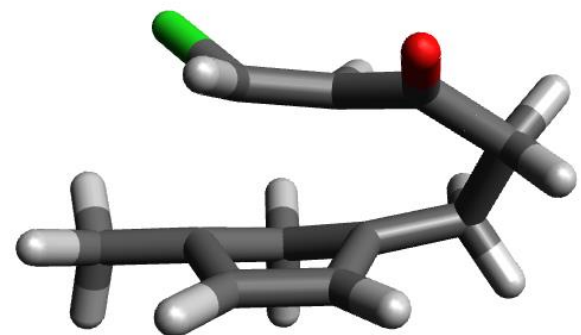


SQM_readfc

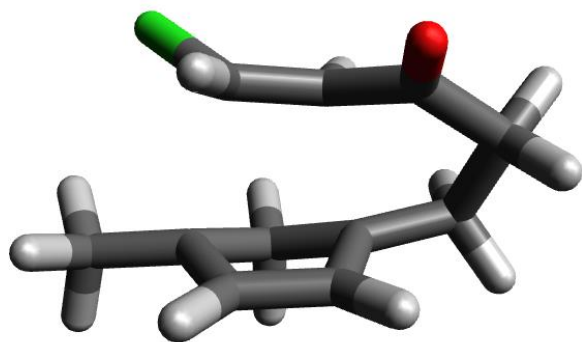
IMDA_1



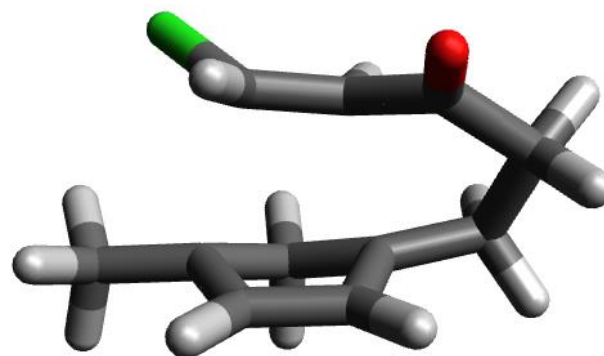
IMDA_1: conformations



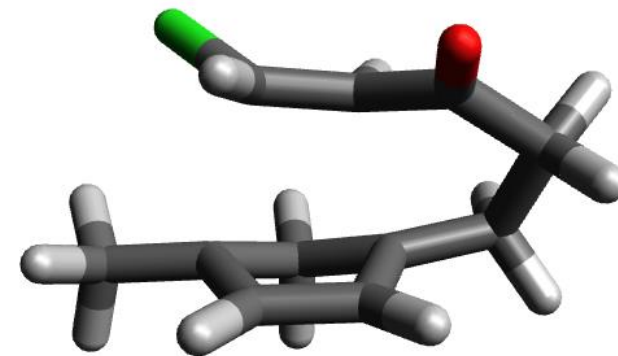
DFT Hessian



HF Hessian

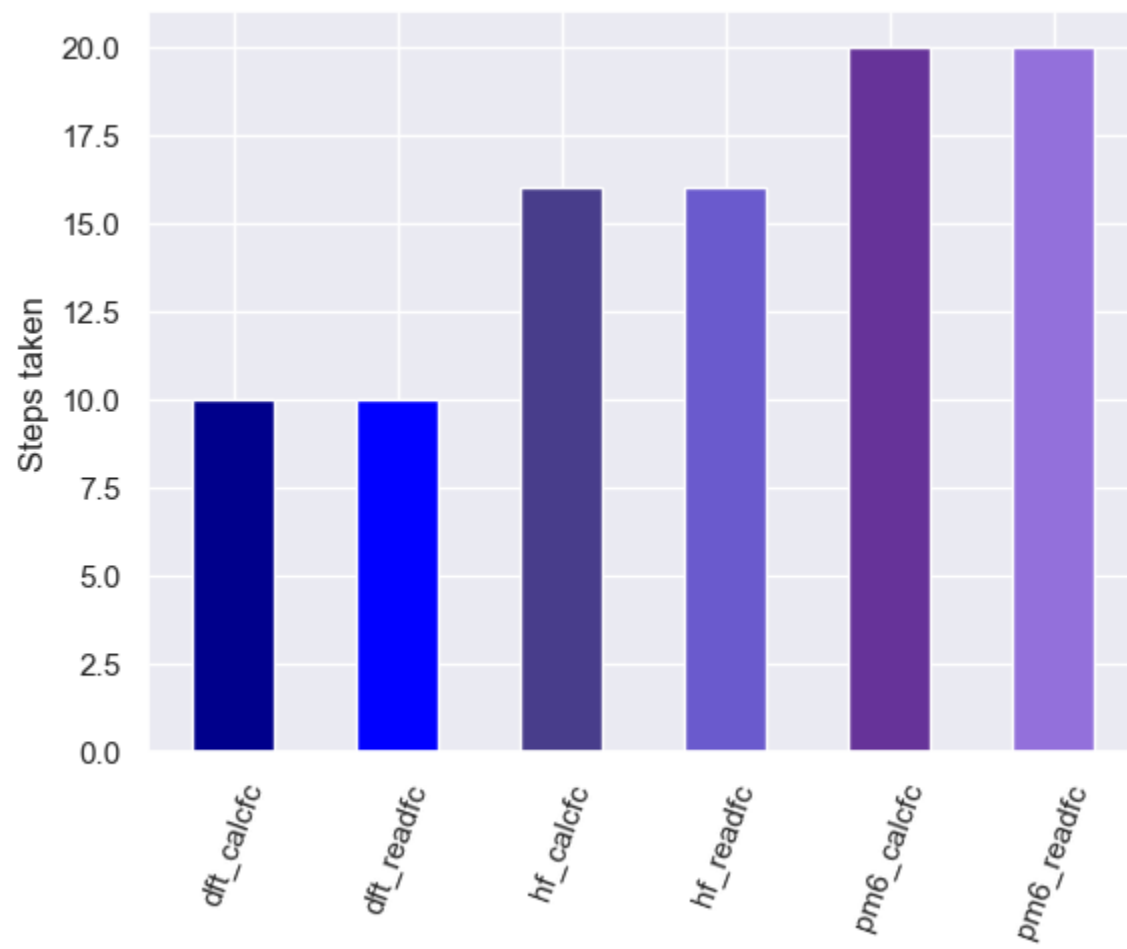
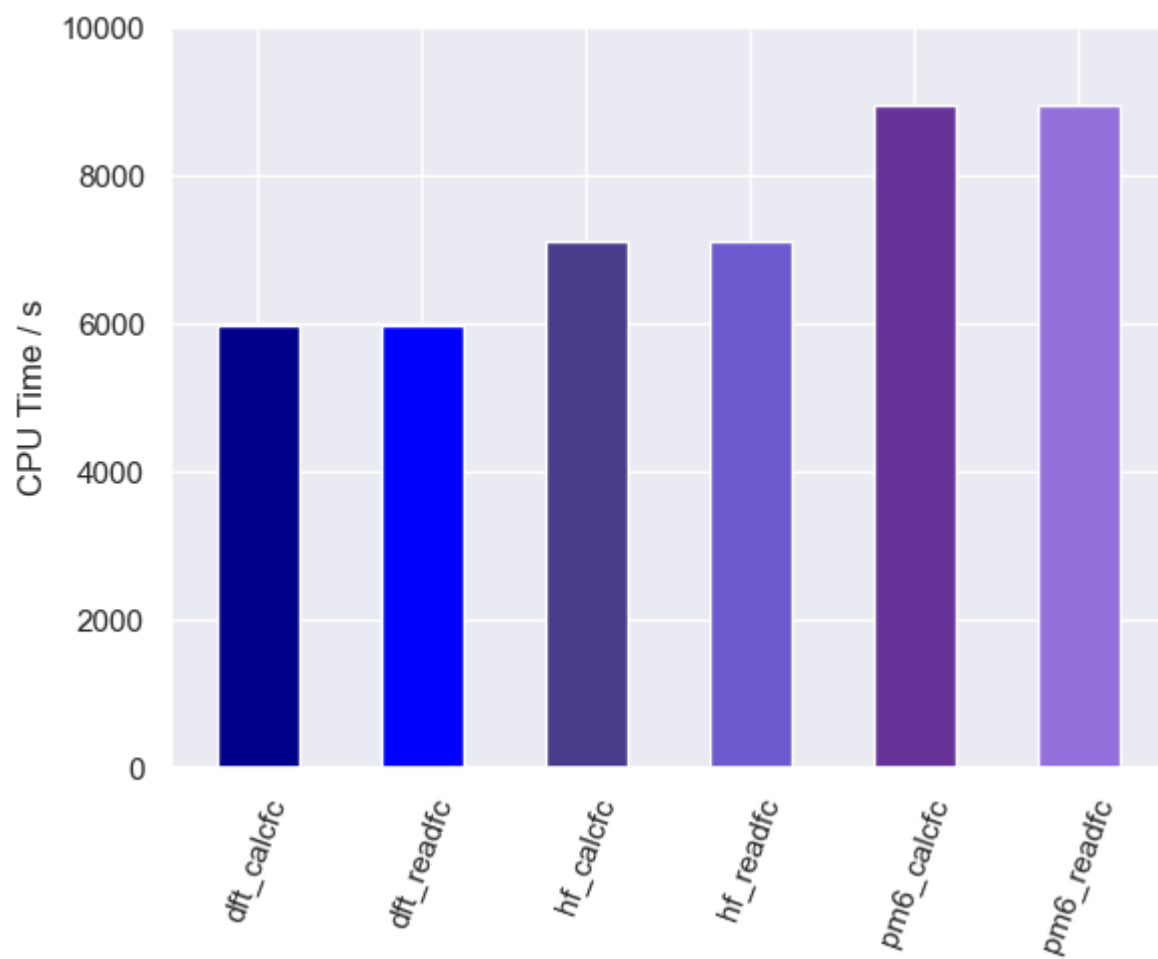


SQM_calcfc

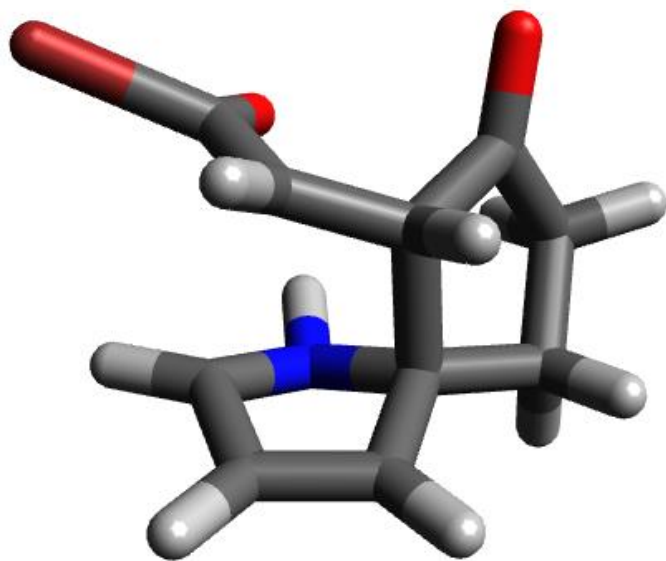


SQM_readfc

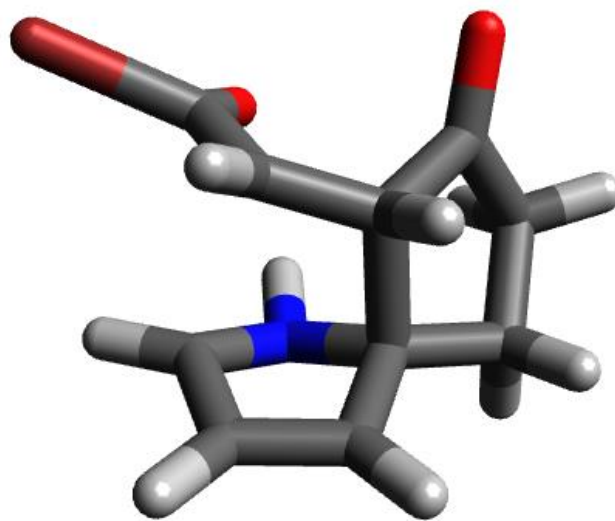
IMDA_2



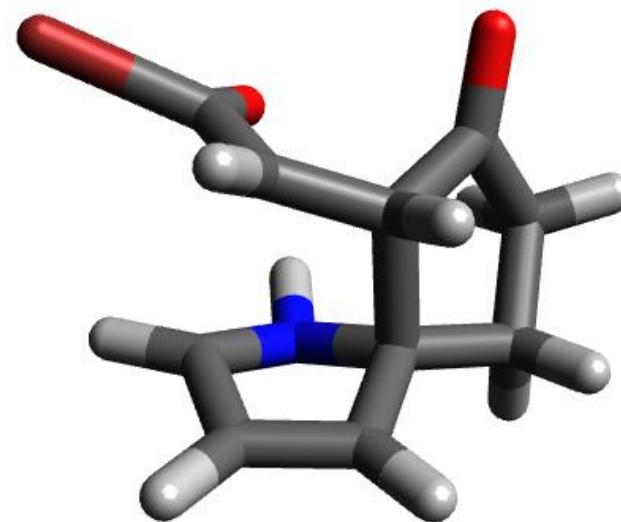
IMDA_2: conformations



DFT Hessian

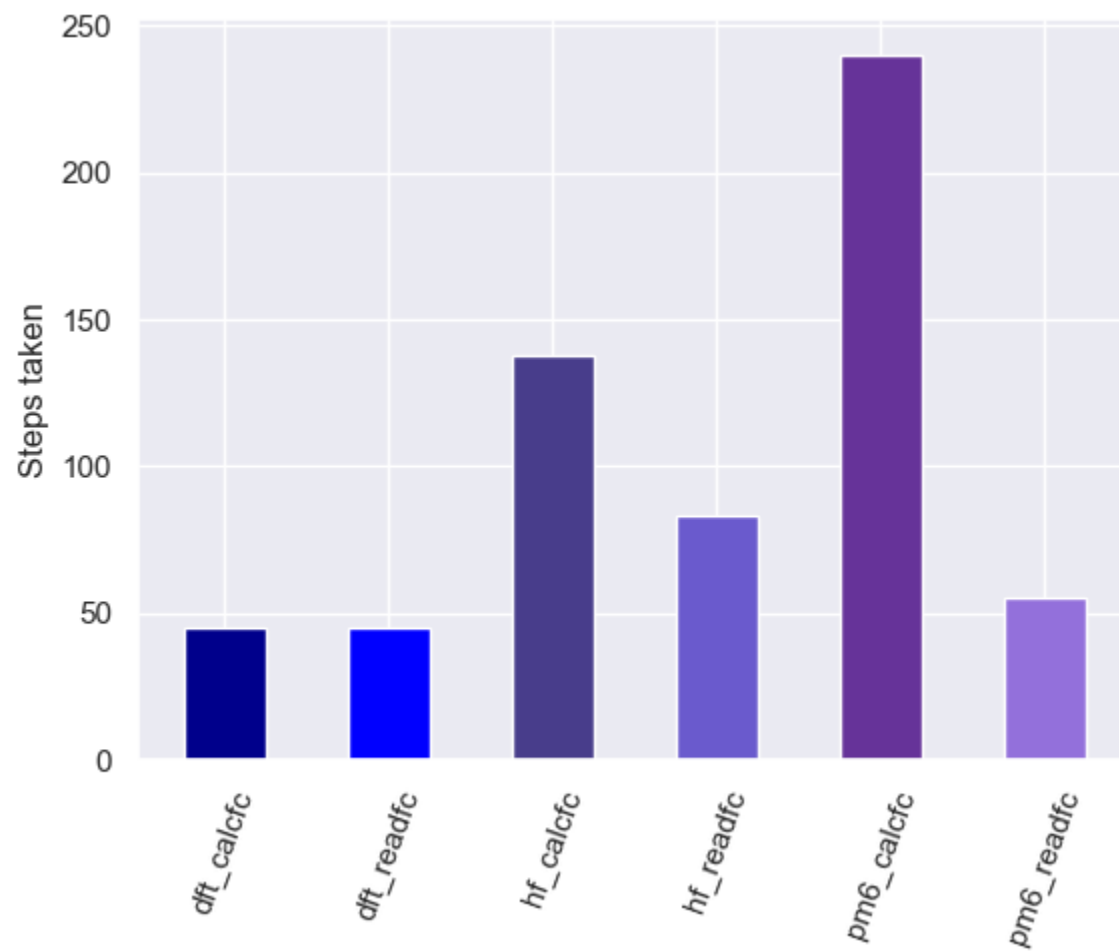
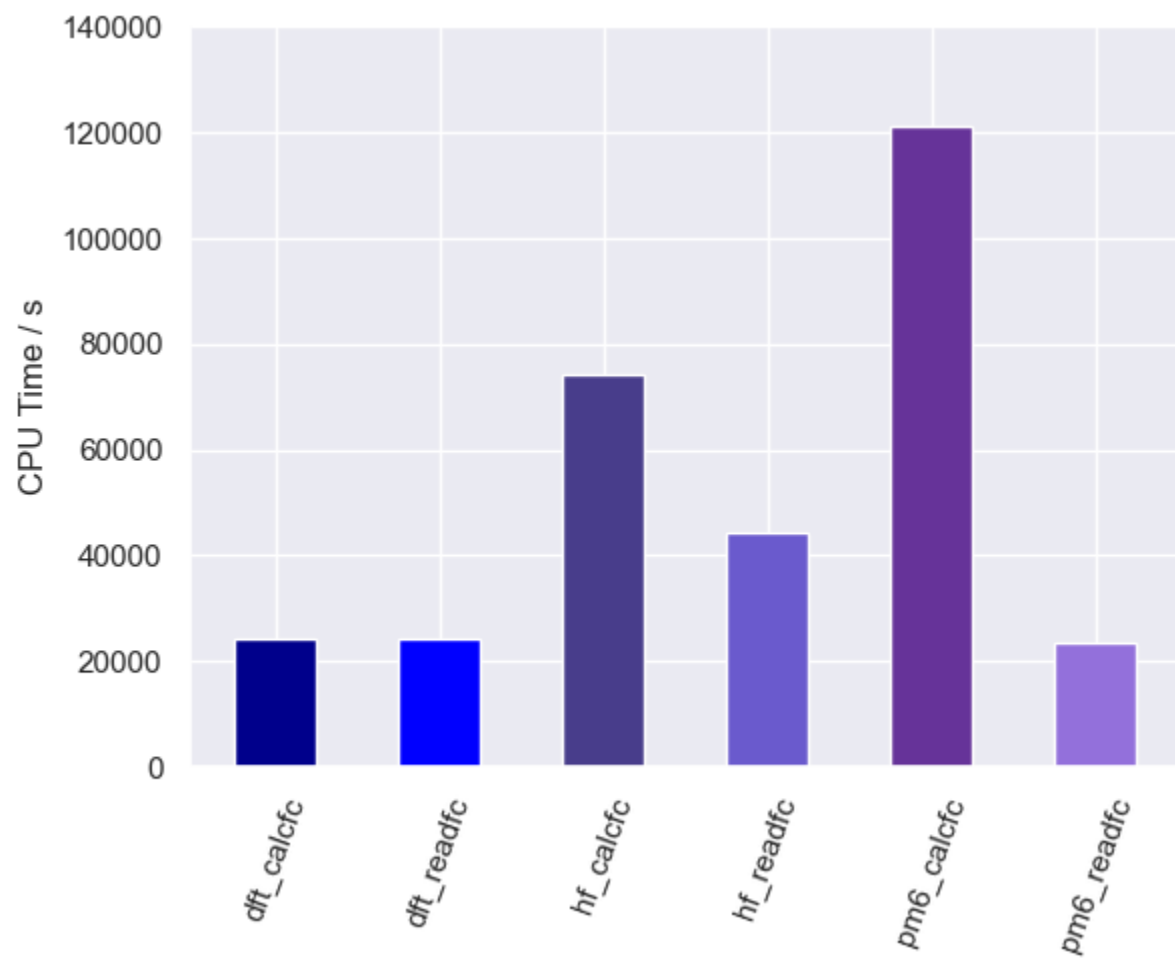


HF Hessian

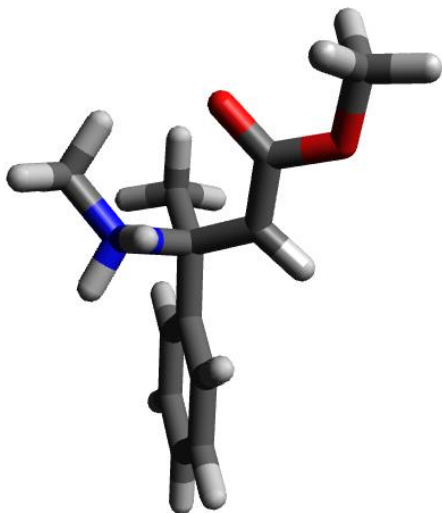


SQM Hessian

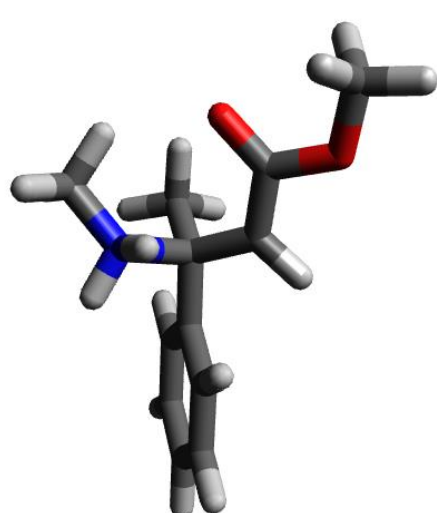
AM_92



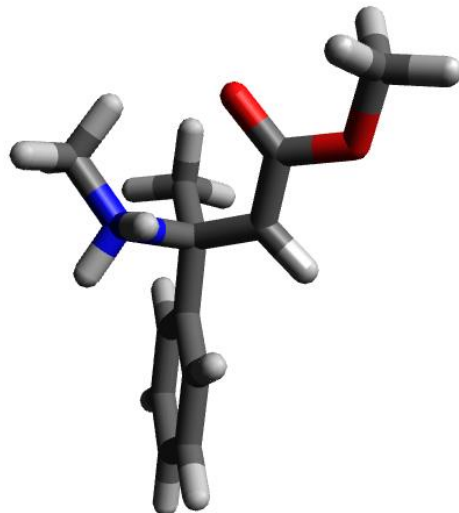
AM_92: conformations



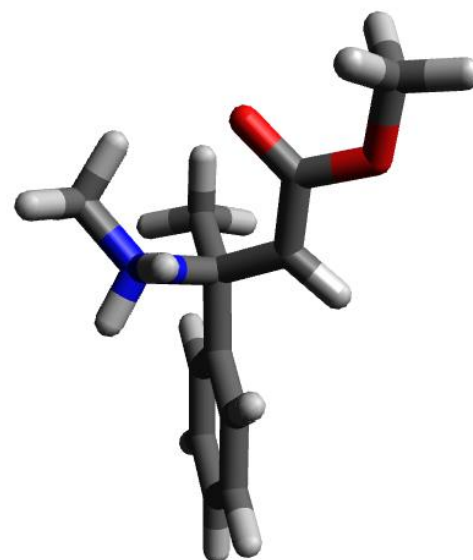
DFT Hessian



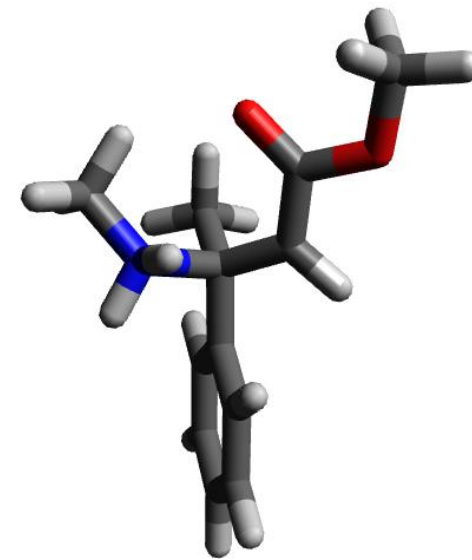
HF calcfc



HF readfc



SQM calcfc

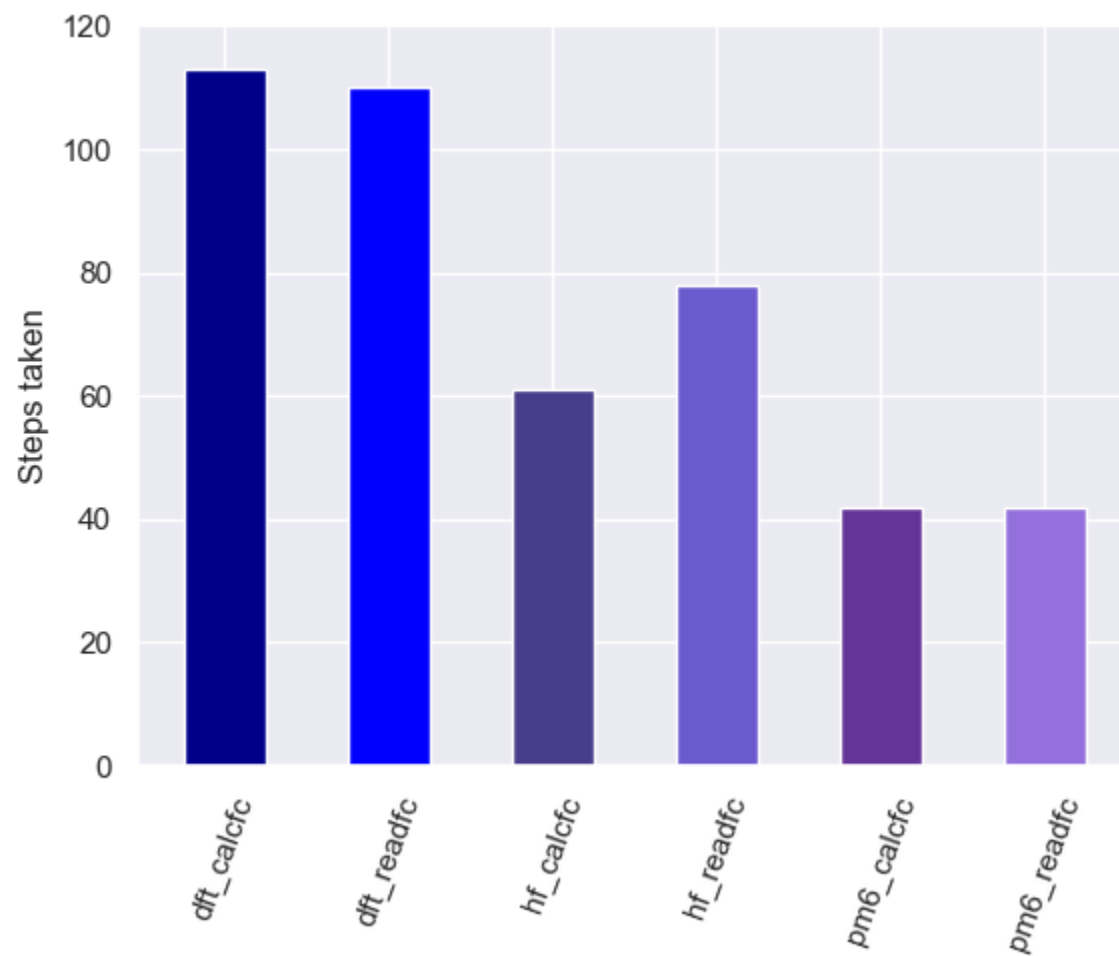
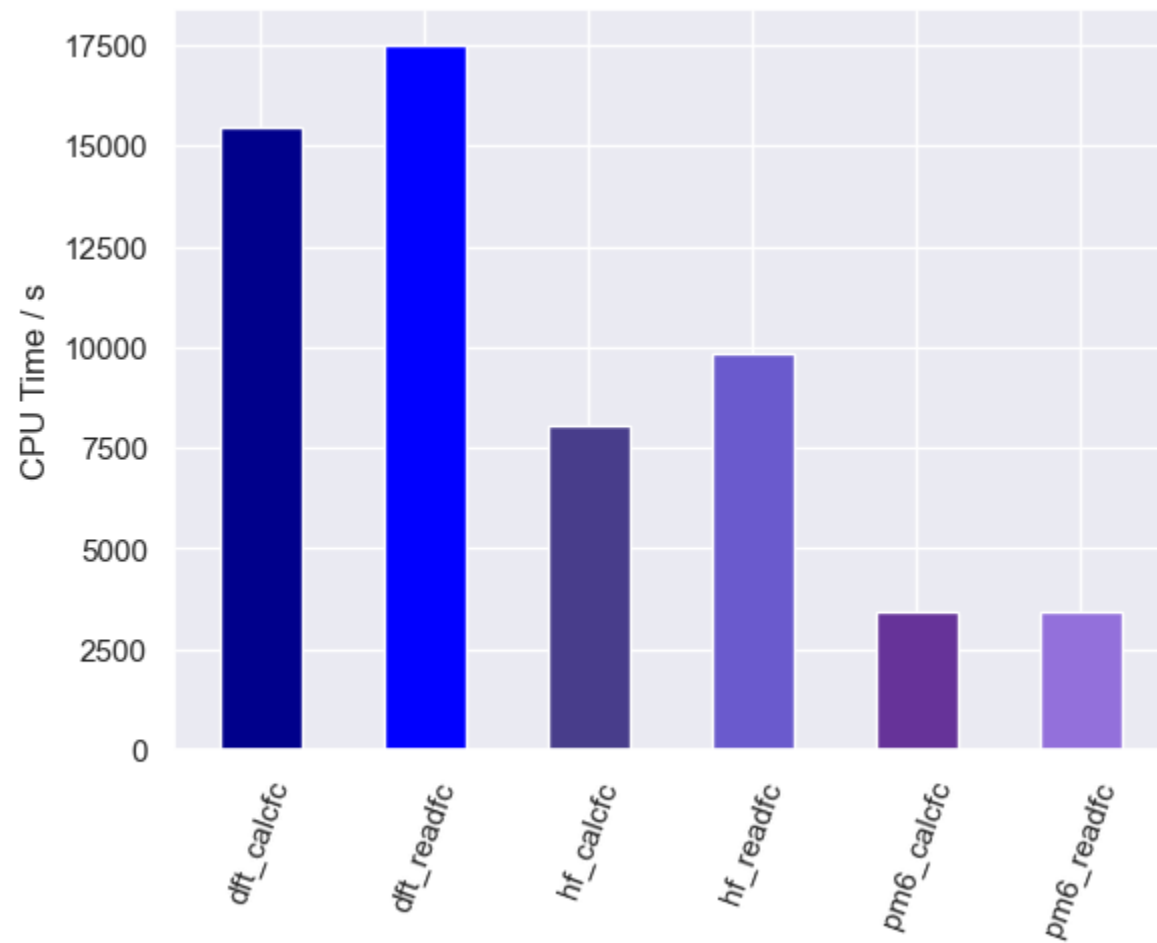


SQM readfc

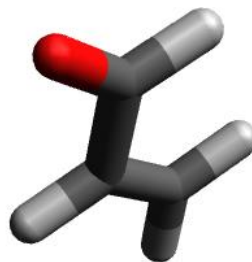
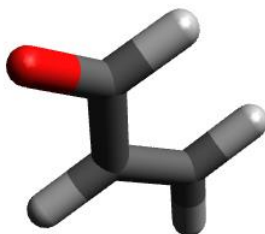
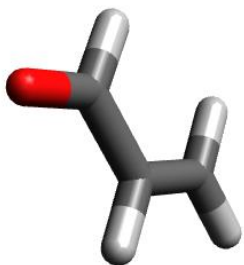
Incorrect TS:

All: N---C reacting bond rotation at around 77 1/cm

DA_1



DA_1: conformations



DFT calcfc

DFT readfc

HF calcfc

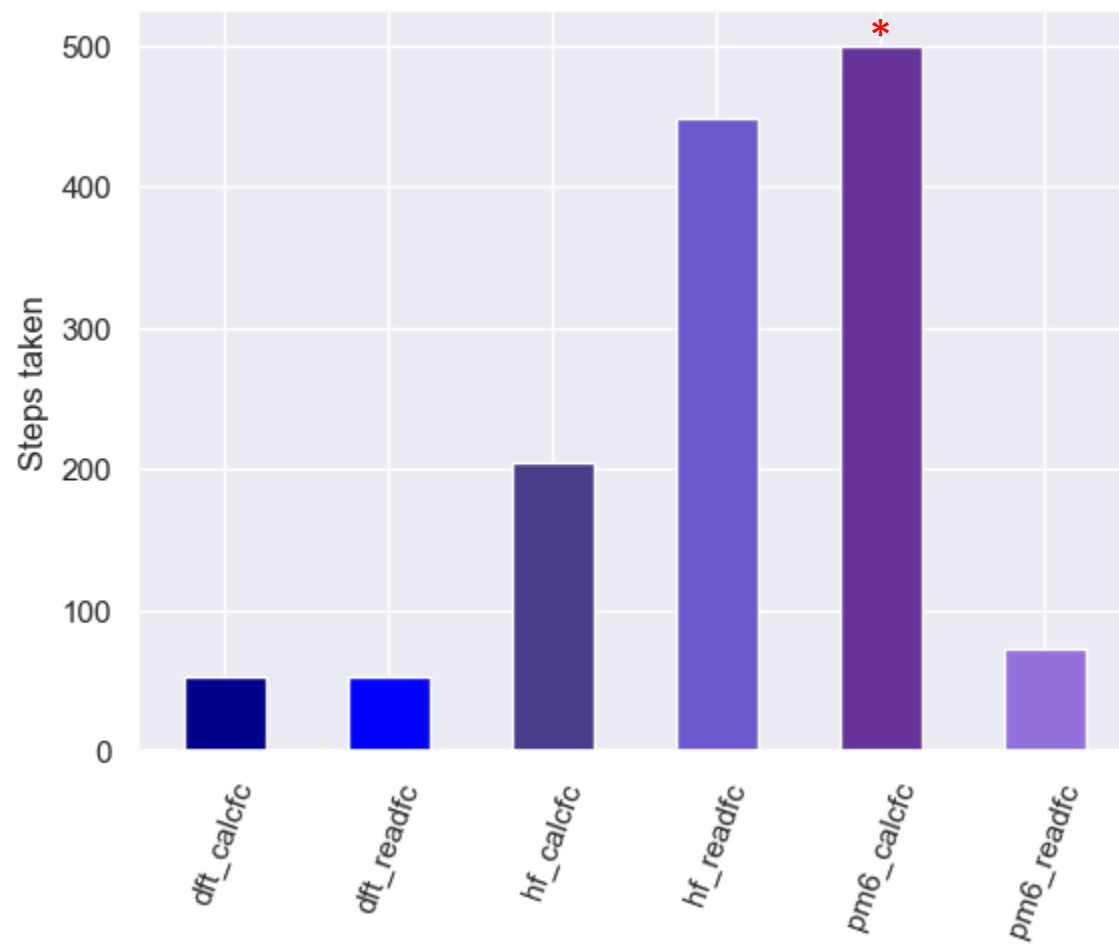
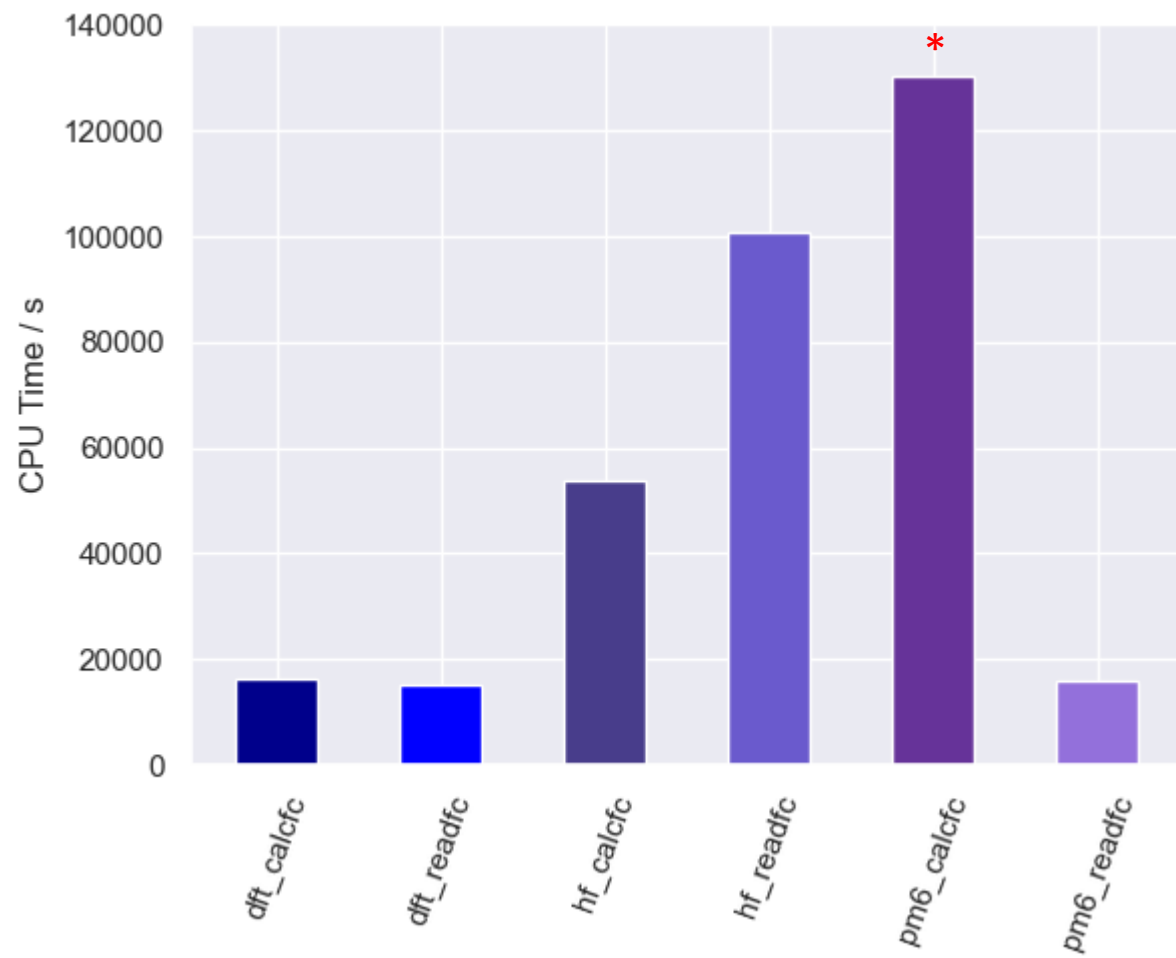
HF readfc

SQM Hessian

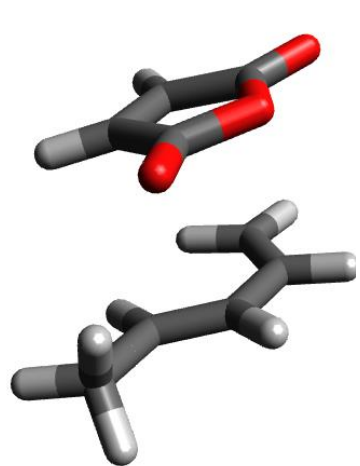
Incorrect TS:

All: Symmetric rocking at around 20-30 1/cm

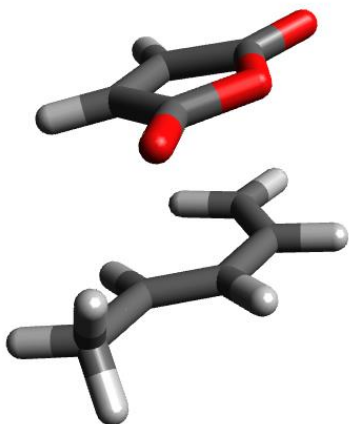
DA_2



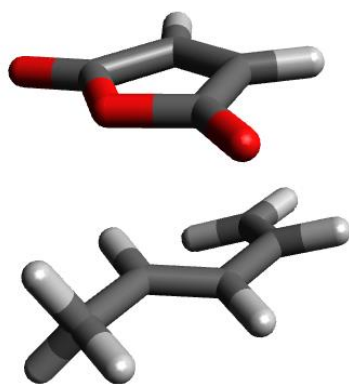
DA_2: conformations



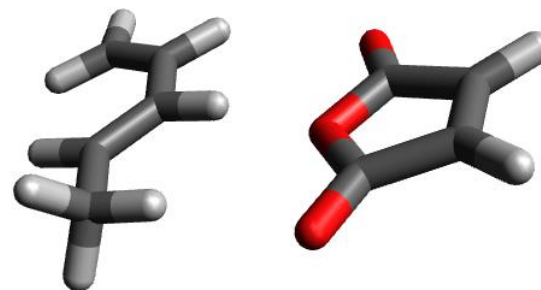
DFT calcfc



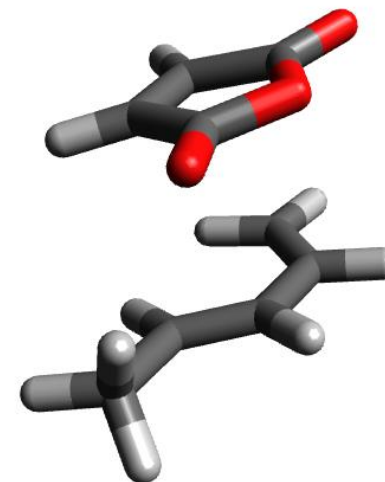
DFT readfc



HF calcfc



HF readfc



SQM readfc

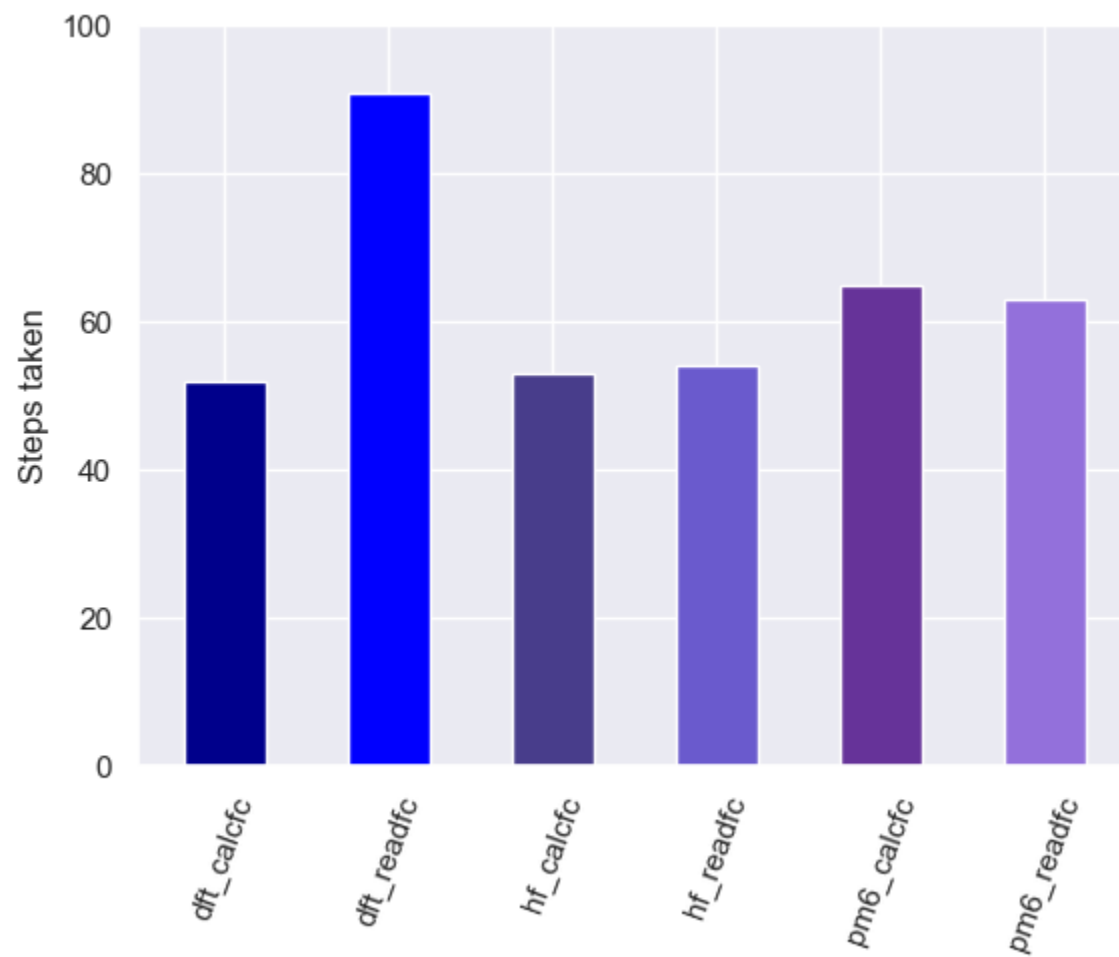
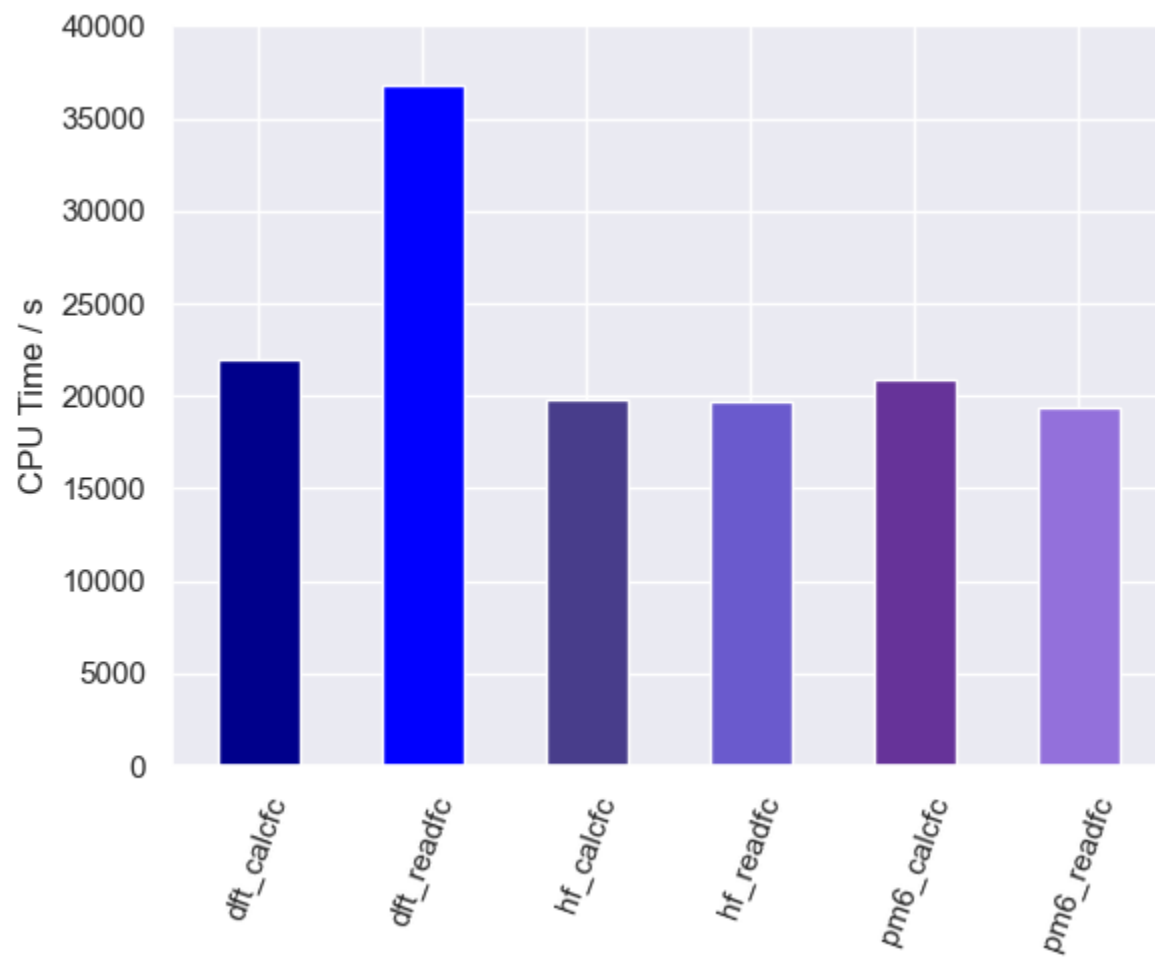
(SQM calcfc did not converge)

Incorrect TS:

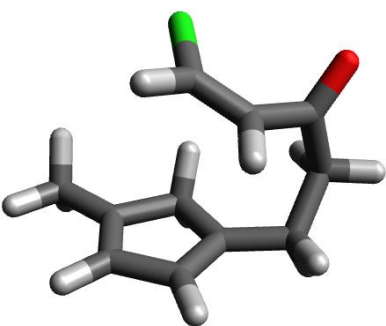
DFT_calcfc, DFT_readfc and SQM_readfc: Anti-Houk to Houk RHC-Me rotation at around 180 /1cm

HF_calcfc, HF_readfc: Diene rocking at around 25-85 1/cm

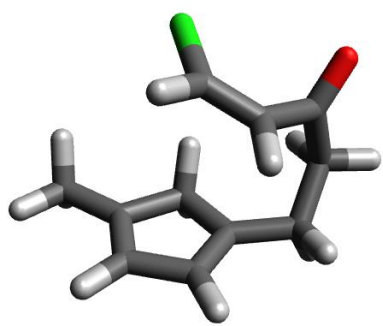
IMDA_1



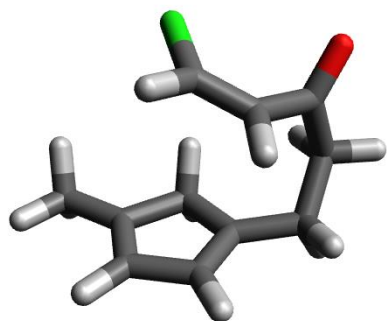
IMDA_1: conformations



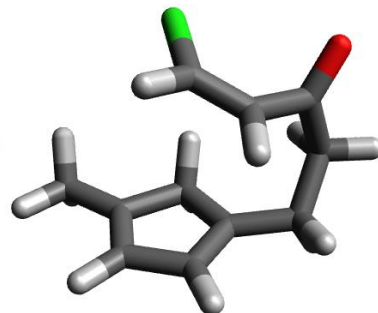
DFT calcfc



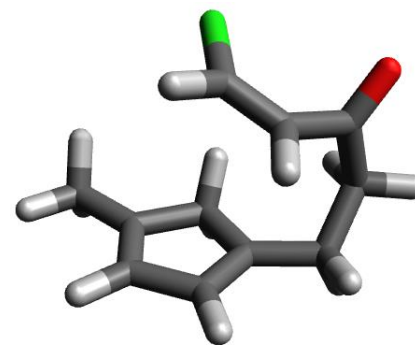
DFT readfc



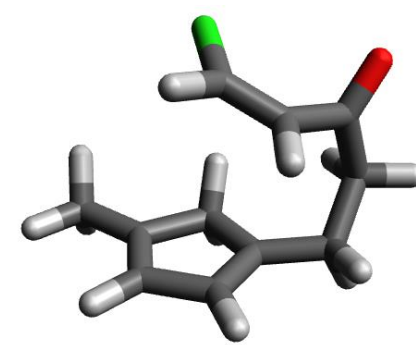
HF calcfc



HF readfc



SQM calcfc

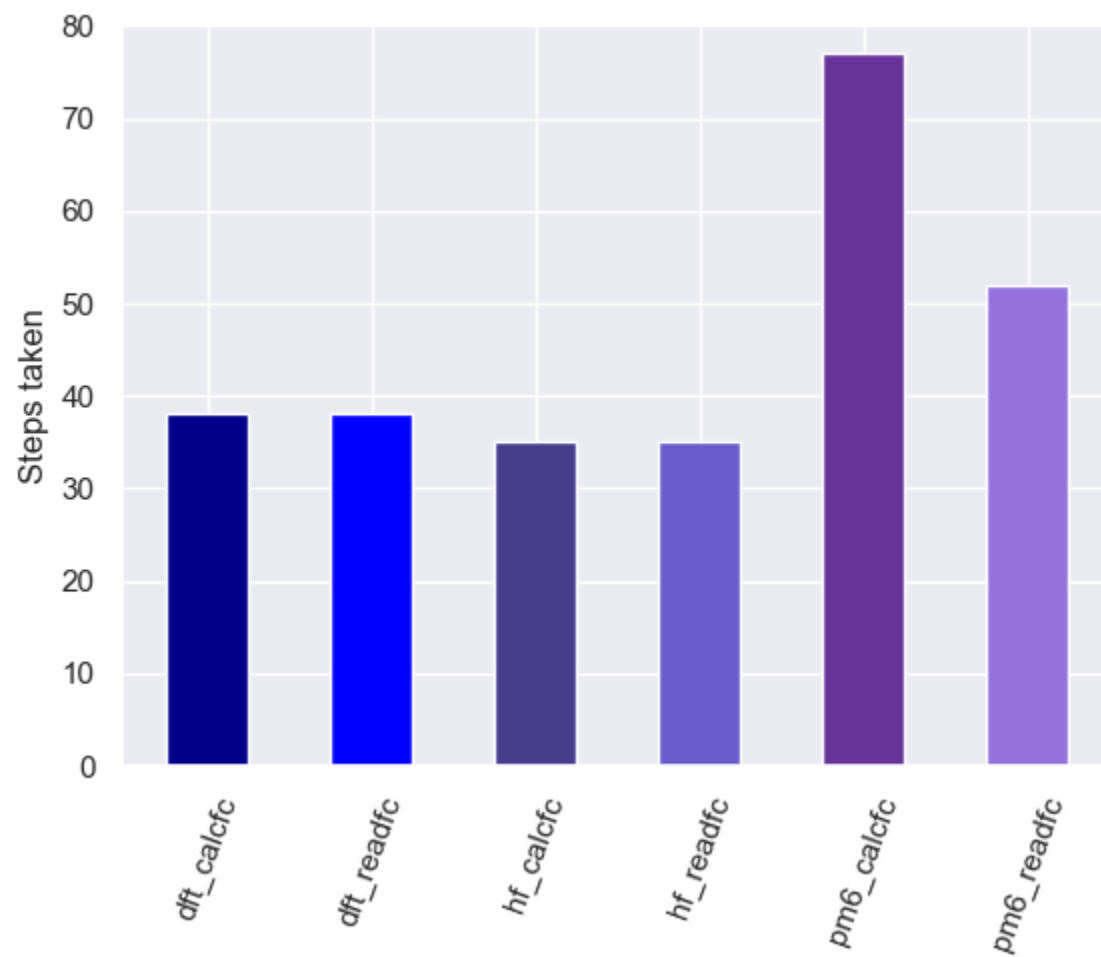
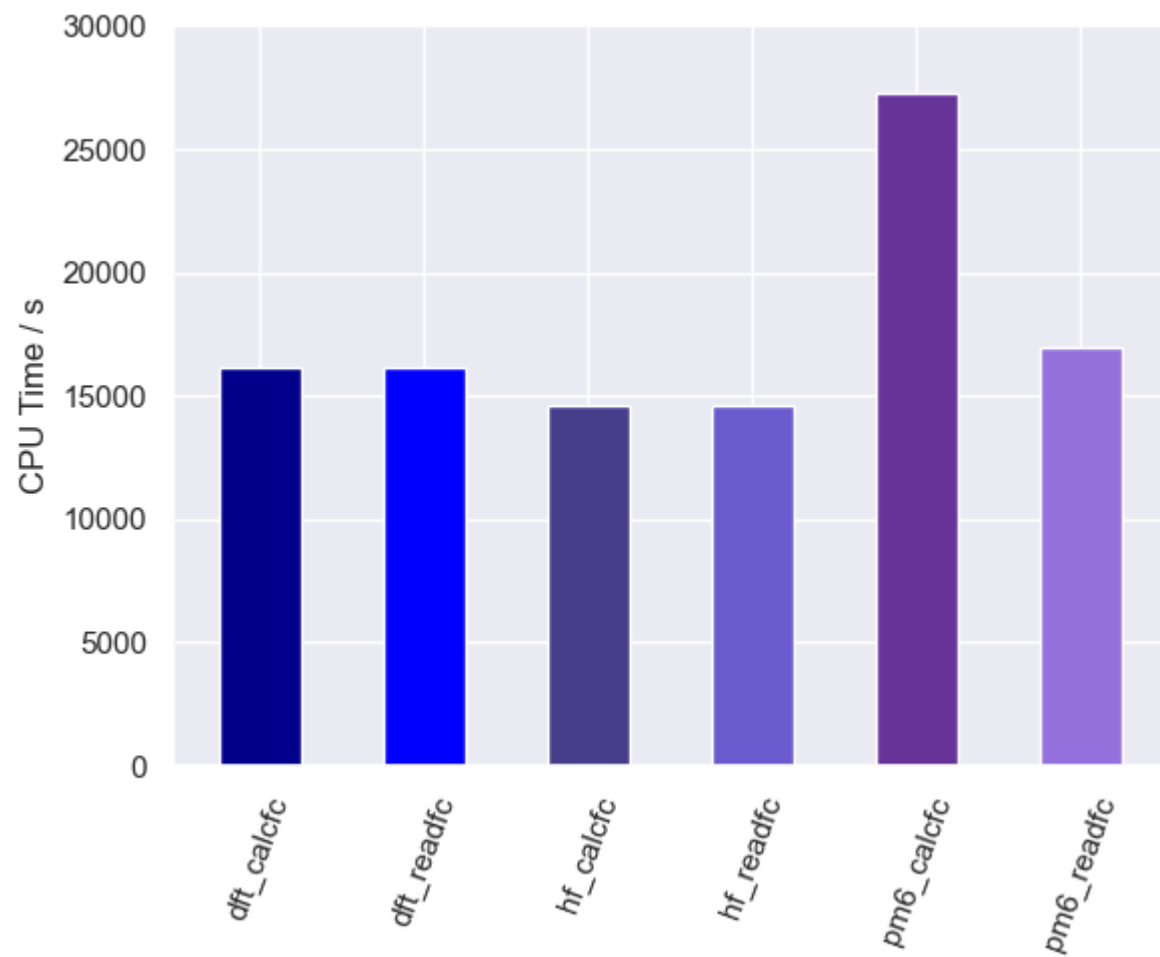


SQM readfc

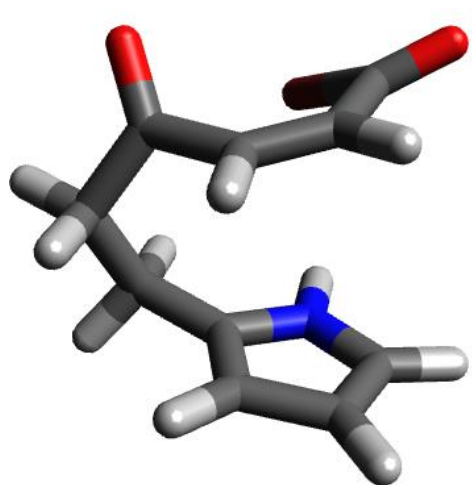
Incorrect TS:

R(O)C-CH₂(R) bond rotation at around 28 1/cm

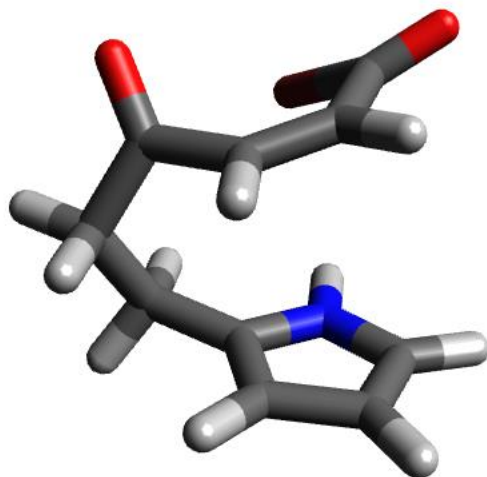
IMDA_2



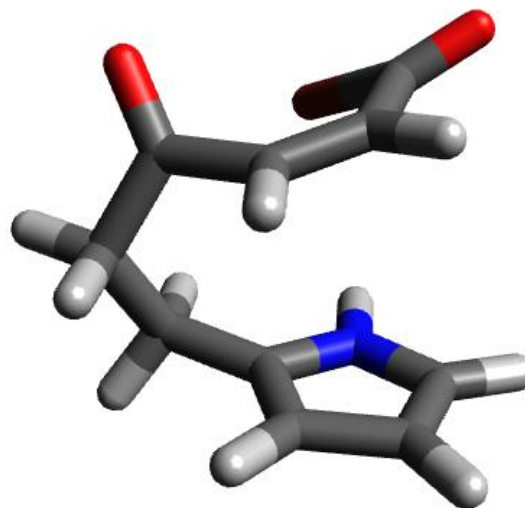
IMDA_2: conformations



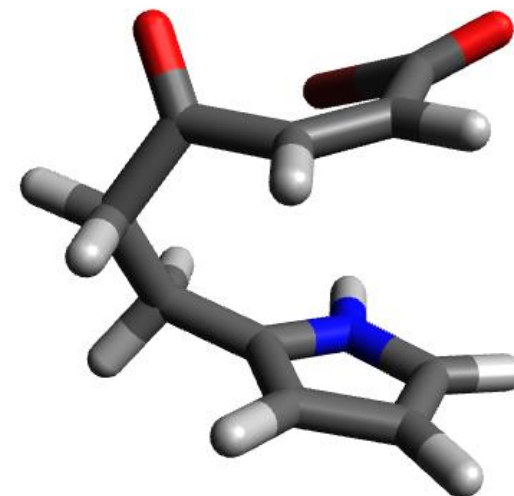
DFT Hessien



HF Hessien



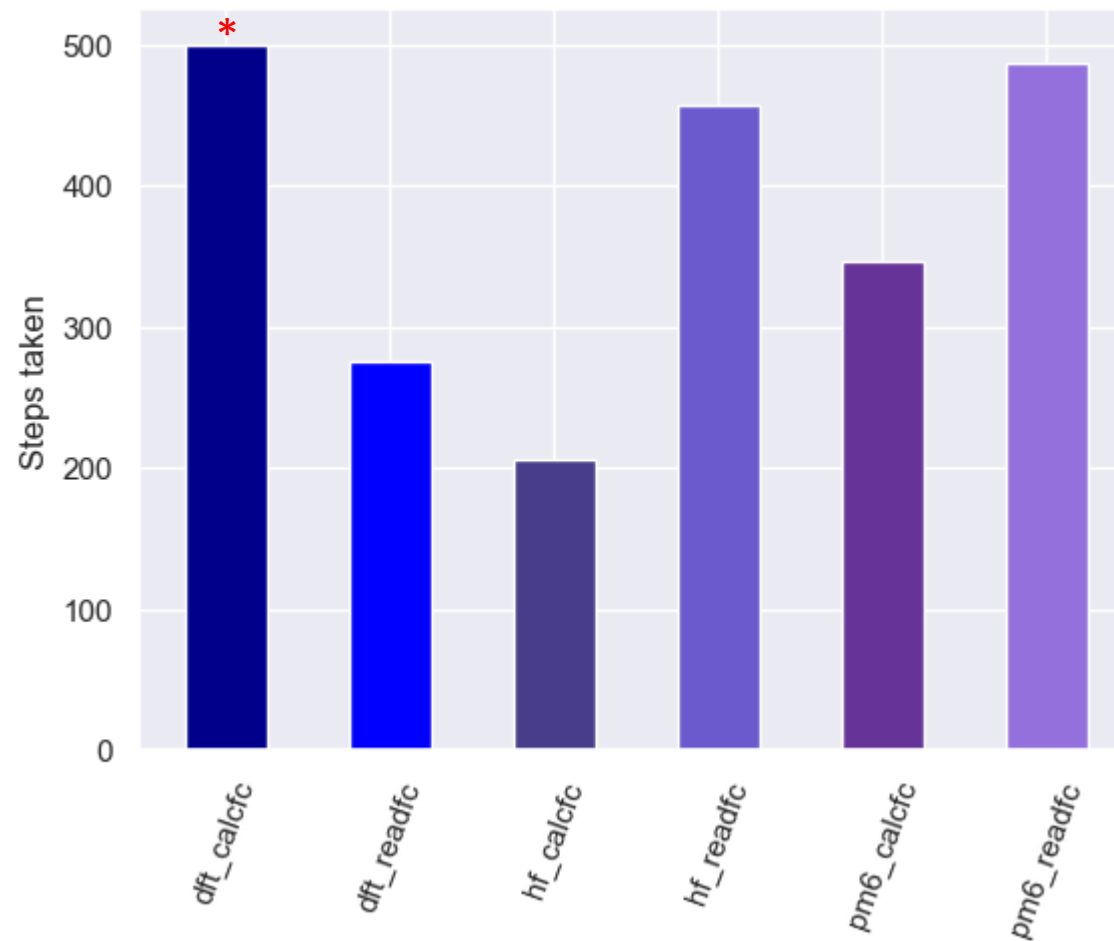
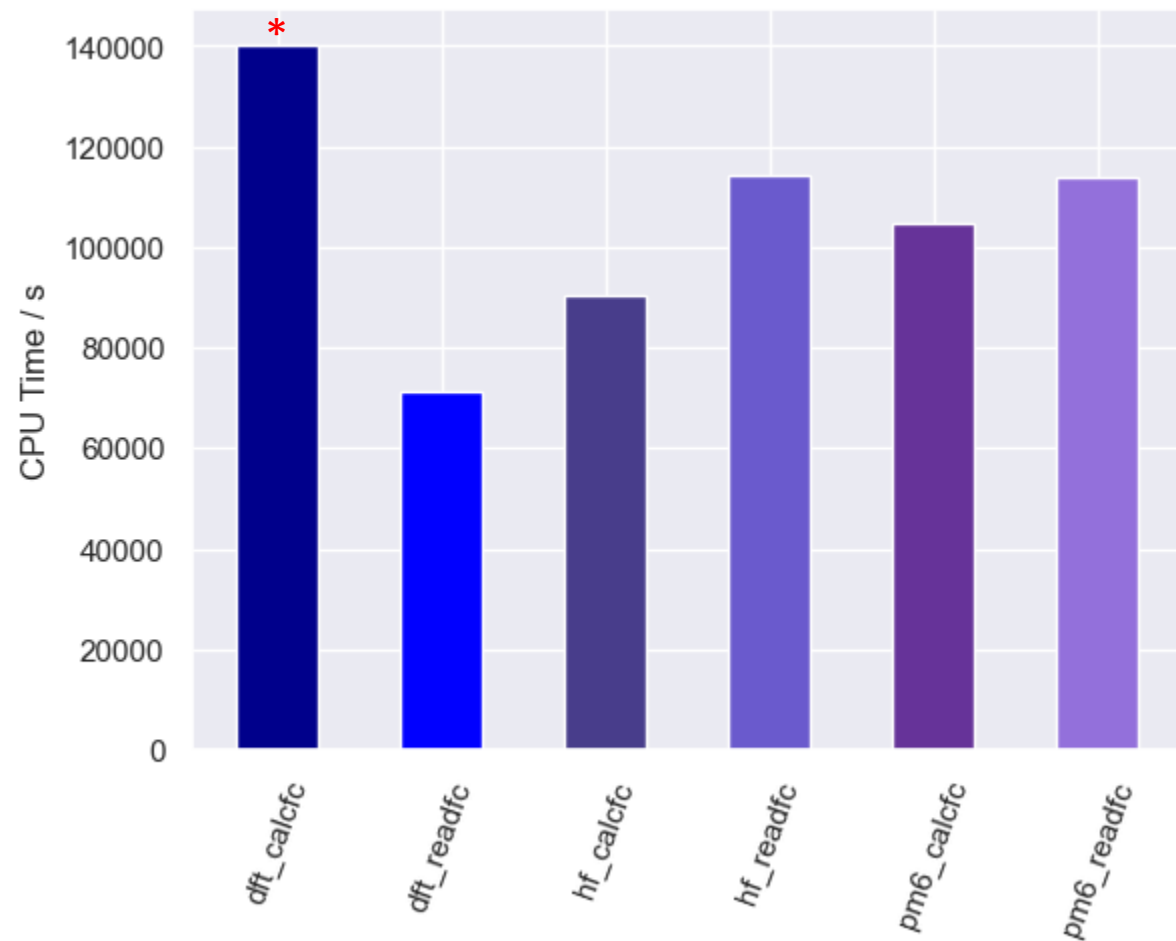
SQM calcfc



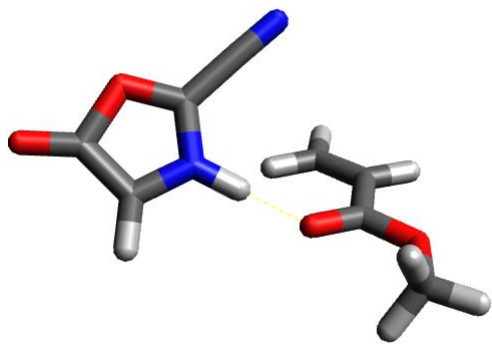
SQM readfc

Incorrect TS:
Br(O)C-CH₂(R) bond rotation at around 10 1/cm

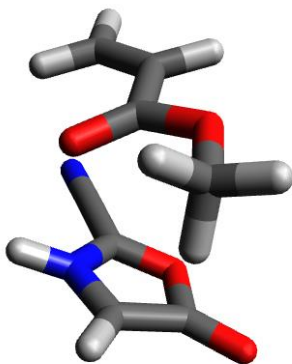
cyclic



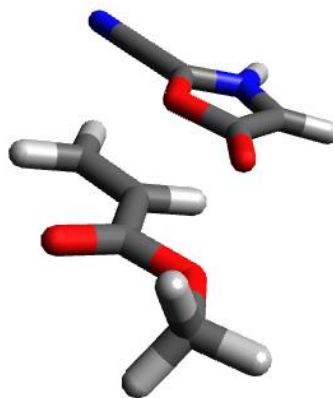
cyclic: conformations



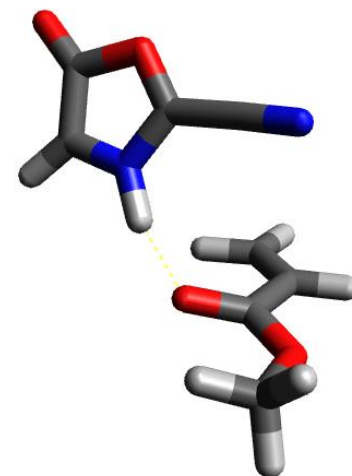
DFT readfc



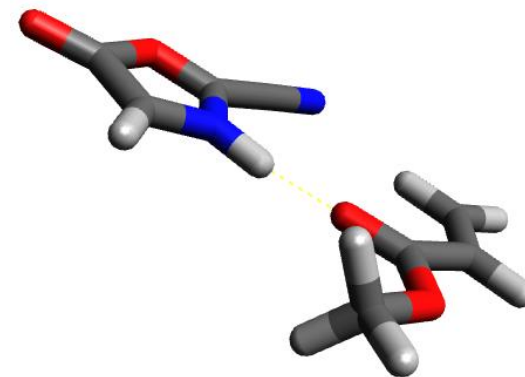
HF calcfc



HF readfc



SQM calcfc



SQM readfc

(DFT calcfc did not converge)

Incorrect TS:

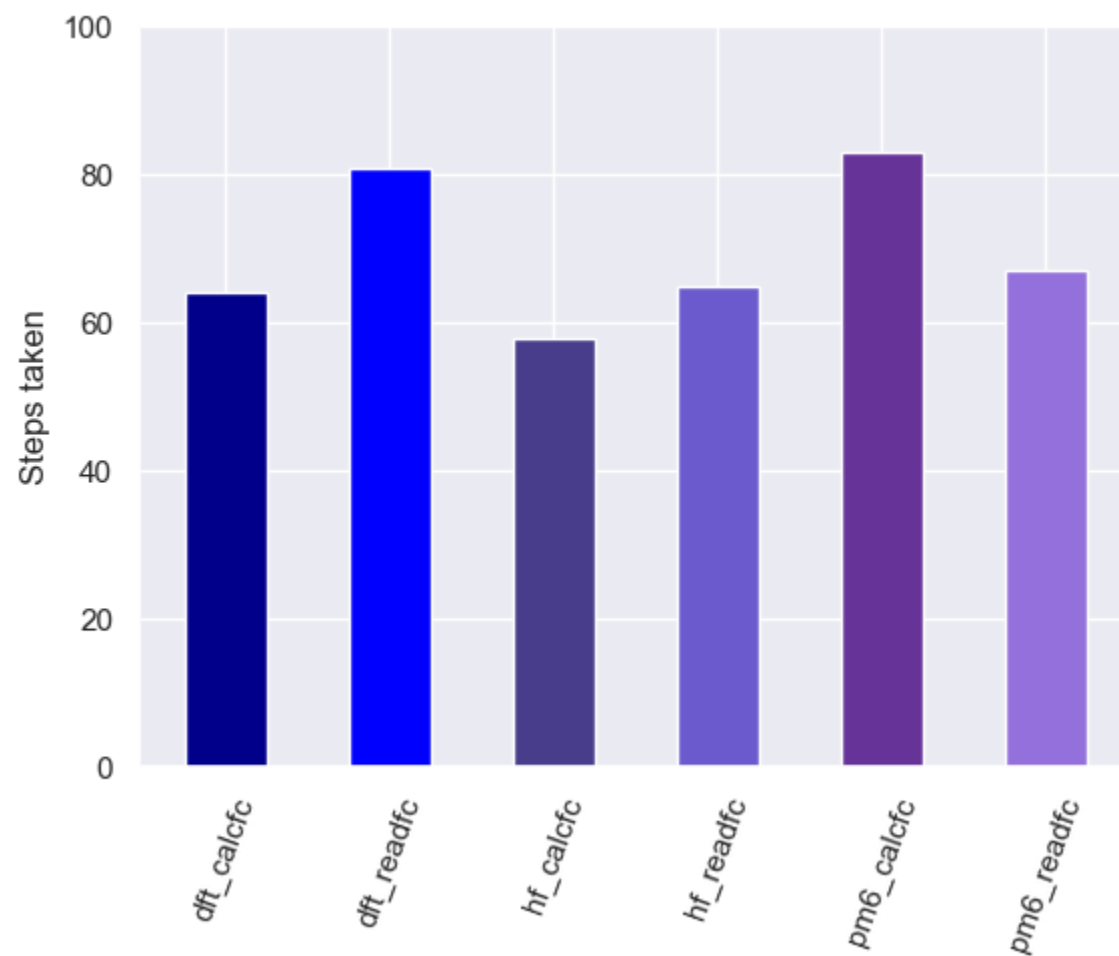
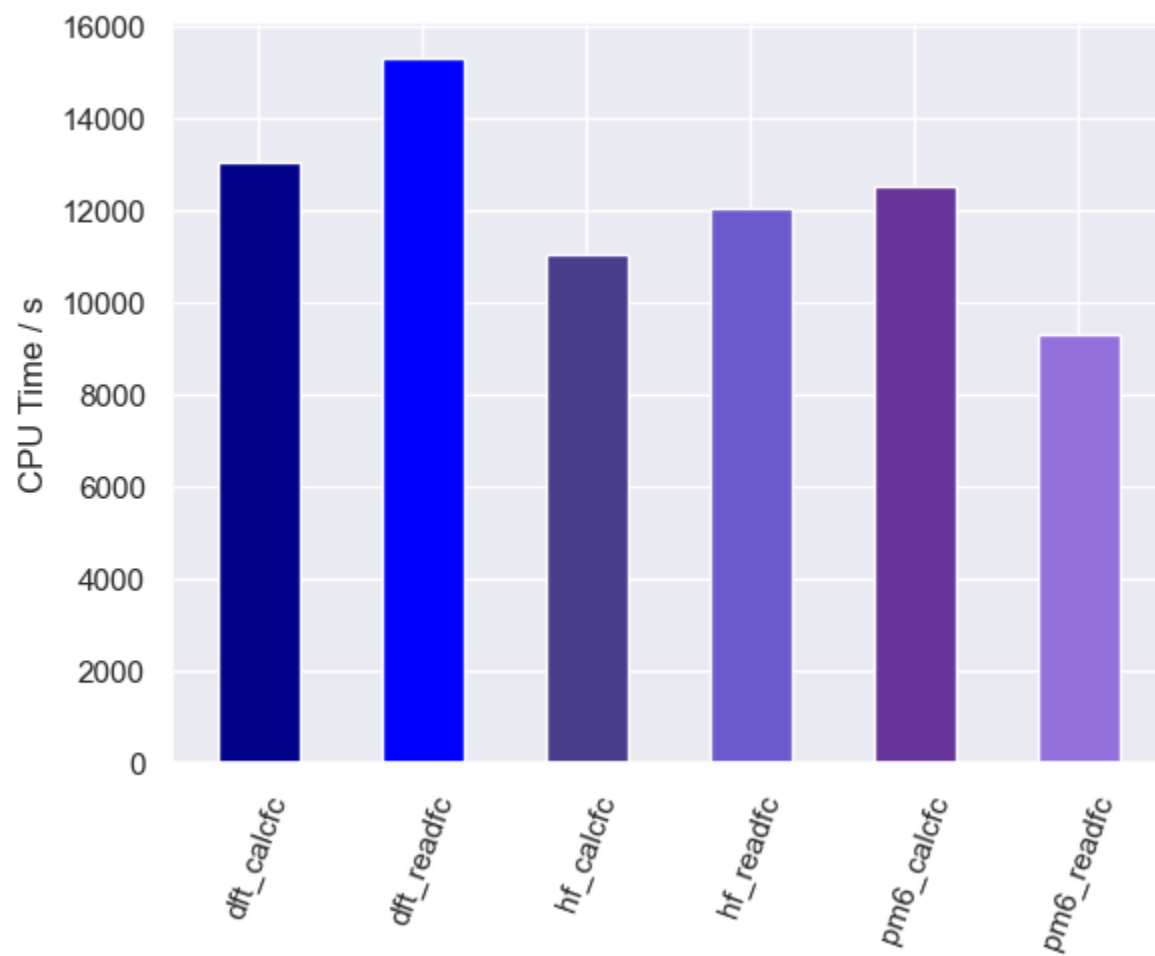
DFT_readfc and SQM_readfc: (R)2NH--O=C(OMe)R H-bond rotation at around 10 1/cm

HF_calcfc: dipole rocking at around 42 1/cm

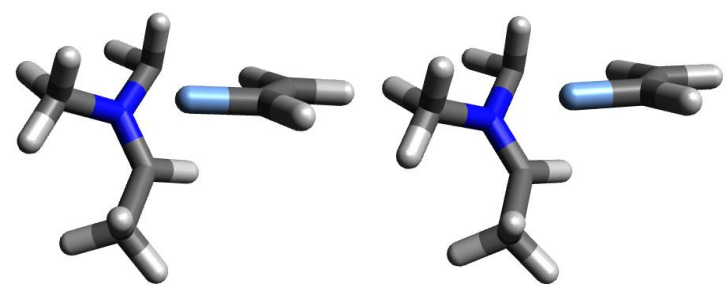
HF_readfc: not a TS (second order saddle point)

SQM_calcfc: R(O)C-OMe bond rotation at around 116 1/cm

allylic

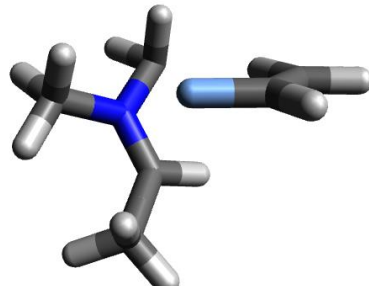


allylic: conformations

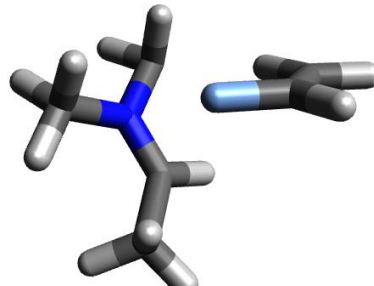


DFT calcfc

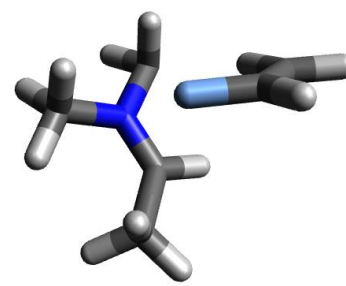
DFT readfc



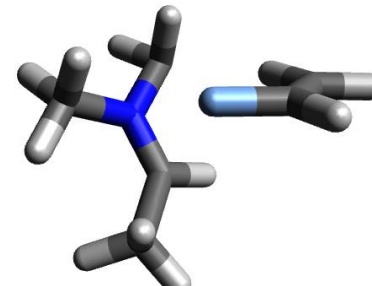
HF calcfc



HF readfc



SQM calcfc



SQM readfc

Incorrect TS:

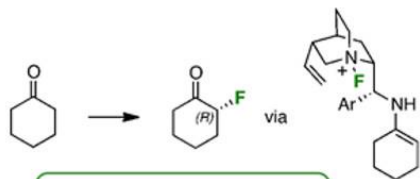
All: (CH₂)-N(CH₃)R bond rotation at around 360 1/cm

Conclusions

- DFT_calcfc and DFT_readfc are usually equally fast (almost none required reoptimisation)
- HF_calcfc and HF_readfc are usually equally fast (almost none required reoptimisation)
- SQM_calcfc is usually similar to SQM_readfc, or slightly slower
- HF initial Hessians were generally the fastest methods, (closely) followed by DFT Hessians and then SQM (PM6) Hessians. This could be because the HF Hessian balances accuracy with calculation speed better than the SQM Hessian, and therefore fewer optimisation steps are needed with a HF Hessian compared to an SQM Hessian
- Computational speedup is not very large compared to the traditional DFT_calcfc method (HF_calcfc is approximately 80% the CPU time of DFT_calcfc, on average)

Alternative project: Human-guided TS conformation ranking by eigenvector analysis

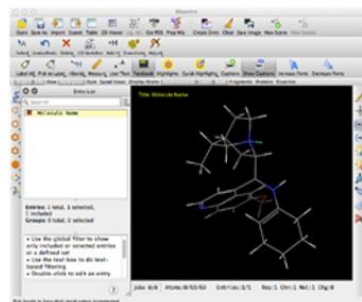
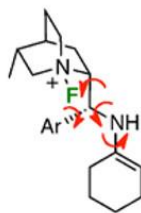
Lam, Y. H., Grayson, M. N., Holland, M. C., Simon, A., & Houk, K. N. (2016). Theory and modeling of asymmetric catalytic reactions. *Accounts of Chemical Research*, 49(4), 750-762.



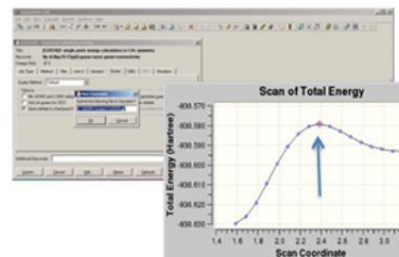
MECHANISTIC HYPOTHESIS

- enamine activation of ketone
- Lewis activation of NFSI ("F⁺")

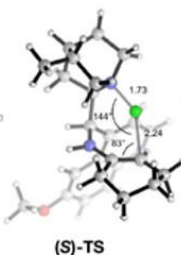
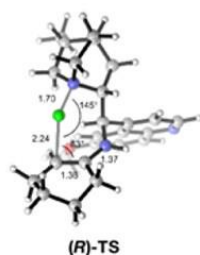
(1) Hypothesis



(2) Conformational Search
(Force Field Calculations)



(3) Locate Transition Structures
(Density Functional Calculations)

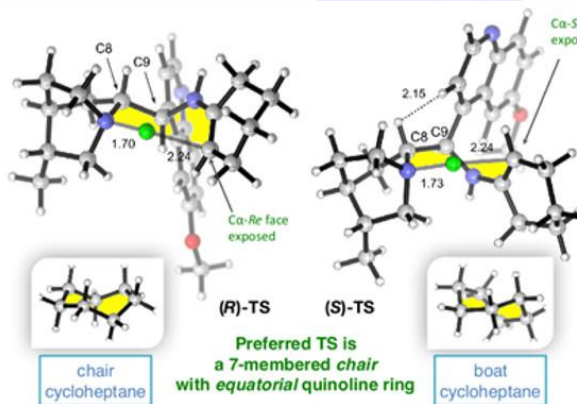


$\Delta\Delta G^\ddagger$ (kcal/mol)

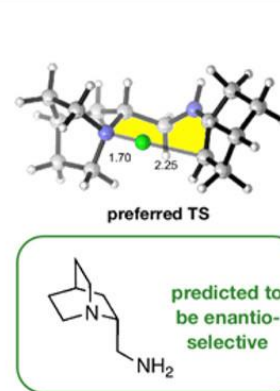
(R)-TS 0
(S)-TS 6.9

computed > 99% ee
experimental 99% ee

(4) Optimize Geometries, Compute Energies,
Evaluate Hypothesis
(Density Functional Calculations)



(5) Acquire Insights, Develop Models



(6) Make New Predictions

(2.5) Conformational ranking/filtering
(Density Functional Frequency Calculations)

- Compute Frequencies with freq keyword (Hessian eigenvalues and eigenvectors)
- Human selects internal coordinates they expect to contribute most to the final TS eigenvector (e.g. C—N bond in Aza-Michael)
- Automatically rank TS conformations by contribution of this internal coordinate to the eigenvector whose eigenvalue is the most negative (this is the eigenvector used in the eigenvector-following algorithm)
- These Hessians can then be used at the start of TS searching (they are required in any case – therefore the only computational cost is in the eigenvector calculations)

